Atmosic MPTool

User Guide

SUMMARY: This document describes the mass production environment to test ATM2/ATM3 and ATM33/e devices on production lines without RF testing equipment.

CONFIDENTIAL



Atmosic™

CONFIDENTIAL | Doc. No. ATM-UGMPT-0059

Table of Contents

Acronyms and Abbreviations **4**
1. Overview **4**
2. Hardware and Software Requirements **5**
2.1 Supported Hardware 5
2.2 Supported Software 6
2.3 Supported OS 6
3. Environment Setup for the APT **6**
4. Environment Setup for Download (DL) Board **6**
4.1 DL Board Versions 6
4.1.1 DL_V6a 6
4.1.2 DL_V6 6
5. Hardware Setup **7**
6. DUT Design Requirements for Manufacturing **9**
7. Software Setup **10**
7.1 Uninstall Any Previous Versions of the MPTool 10
7.2 Install Atmosic MPTool 11
7.3 Atmosic Production Tool Shortcuts 13
7.4 Install MFx Interface 14
7.5 Uninstall MFx Interface 17
8. Adjust Serial Port Latency Timer **17**
9. GUI Mode Quick Start **18**
9.1 Select ATM File 18
9.2 RF Testing with APT 19
9.2.1 Test Items 19
9.2.2 Testing Options 20
Choose Chip Family 20
UART0 Pin Modification 20
10. Atmosic Mass Production Tool User Interface **20**
10.1 Menus 22
10.1.1 File Menu 22
10.1.2 Setting Menu 23
10.1.3 Help Menu 25
10.2 Run Tab 25
10.3 Settings Tab 29
10.4 Crystal Trim 30
10.5 Frequency Test 32
10.6 TX Output Power 32
10.7 RX Sensitivity 33
10.8 Function Test 35
10.9 User Firmware (.atm, .mpbin, .elf) / NVDS file (.bin) / Tags Customization 38
10.10 OTP File (.nvm) (ATM2/ATM3 only) 42
10.11 UART0 Pin Modification 43
10.12 Start Option 44

CONFIDENTIAL

10.13 Write MP Region (This is only supported for ATM2/ATM3) 45
10.14 Command Mode 46
10.15 PV Harvesting Test 46
10.16 External NVDS Process 48
10.16.1 Get External NVDS Process 49
10.16.2 Notify External NVDS Used Process 49
10.17 Update Test Result with External Process 50
10.18 Calibration Settings 52
11. Atmosic Production Tool Console Mode 52
11.1 Windows System Environment Variable 53
11.2 Board Discovery 53
11.3 DUT/AG Tx Test 54
11.4 DUT/AG Rx Test 54
11.5 DUT/AG Tx Power Gain 55
11.6 DUT RAM or OTP Programming 55
11.7 DUT Firmware Programming 56
11.8 DUT NVDS Programming or Pull 57
11.9 DUT Firmware Validation 57
11.10 DUT Firmware and NVDS Programming Concurrently 58
11.11 DUT Flash Erase 59
11.12 DUT/AG Reset Command 59
11.13 DUT/AG Version 59
11.14 DUT/AG NVDS Read 60
11.15 DUT Log 60
11.16 Bluetooth LE RF PER Test (Packet Error Rate) 60
11.17 Bluetooth LE RF FOS Test (Frequency Offset) 61
11.18 Bluetooth LE RF CAL Test (Calibration) 61
11.19 OTP/NVDS/MP Region File Content Display 61
11.20 NVDS/MP Region Create Blank File 62
11.21 OTP/NVDS/MP Region Tag Addition with File 62
11.22 OTP/NVDS/MP Region File Merge 63
11.23 DUT/AG Test ADV 64
11.24 DUT/AG Test WuRx 64
11.25 DUT/AG I/O Setting 65
11.26 DUT/AG Set/Get Runtime Crystal CAP 65
11.27 Set UART Rx Enable 66
11.28 Change the UART Pin Mux 66
11.29 Set 16 MHz Crystal Capacitor 67
11.30 Set Testing Log Refresh 67
11.31 Set Testing Log Finished 67
11.32 CLI Detail Log 68
11.33 Write MP Region 68
11.34 Pull MP Region 69
11.35 Set UART Baud Rate 69
11.36 Dump Flash to Bin File 70

CONFIDENTIAL

11.37 Flash Chip Erase 70
11.38 Check Flash Protection (Only for ATM2/ATM3) 70
11.39 Show ATM File Information 1
11.40 Export FW Bin File from ATM File 71
11.41 Export NVDS Bin File from ATM File 71
11.42 FTDI Pin Control 72
11.43 DUT RRAM Firmware Programming (Only For ATM33/e) 73
11.44 DUT RRAM NVDS Programming (Only For ATM33/e) 73
11.45 DUT Secure Journal Dump or Push (Only For ATM33/e) 74
11.46 DUT RRAM Erase (Only For ATM33/e) 74
11.47 DUT Get PV Count 75
12. Multiple DUT Testing in CLI Mode 1
13. Power Control 76
13.1 Using the MPTool GUI 76
13.2 Using the MPTool Console Mode 76
13.2.1 Using Download Board 76
13.2.2 Using APT Board 76
14. NVDS Tool 76
14.1 DISPLAY Content of NVDS and OTP file 77
14.2 Add, Erase Tag 78
14.3 Generate New OTP/NVDS/MP Region File 80
15. Common OTP/NVDS/MP Region Tag ID 81
16. Troubleshooting 82
Revision History 82

CONFIDENTIAL

List of Figures

Figure 2-1 Atmosic Production Tester Kit
Figure 3-1 Atmosic Production Tester Board Configuration
Figure 3-2 Connection Setup Between PC, APT and DUT
Figure 4-1 Production Download Board (DL) Configuration
Figure 4-2 Connection Setup Between PC, DL and DUT
Figure 4-3 DL_V6a Critical Jumper Setting
Figure 4-4 DL_V6 with 2.5 V I/O Voltage Critical Jumper Setting
Figure 4-5 DL_V6 with 1.8 V I/O Voltage Critical Jumper Setting
Figure 5-1 Test Point Connector Pin Definition
Figure 7-1 Uninstall Previous MP Tool Version Completed
Figure 7-2 Atmosic MP Tool Destination Folder
Figure 7-3 Atmosic MP Tool Setup Completed
Figure 7-4 Start Menu Shortcuts
Figure 7-5 COM Port Setup in Device Manager

Figure 7-6 Atmosic MFx driver
Figure 7-7 COM Port Assignment
Figure 8-1 COM Port Setup in Device Manager
Figure 8-2 USB Serial Port Properties
Figure 8-3 Advanced Setting for COM
Figure 9-1 Select ATM File
Figure 9-2 Choose Test Options
Figure 9-3 Burn_RAM Process
Figure 9-4 Choose Atmosic Device
Figure 9-5 Select ATM File
Figure 10-1 Atmosic Mass Product Tool User Interface
Figure 10-2 MP Tool Run Menu
Figure 10-3 MP Tool File Menu
Figure 10-4 MP Tool Setting
Figure 10-5 MP Tool Help Menu
Figure 10-6 Run Tab Before Testing
Figure 10-7 Run Tab Under Testing
Figure 10-8 Run Tab After Testing Success
Figure 10-9 Run Tab After Testing Fail
Figure 10-10 Setting Tab
Figure 10-11 Crystal Trim Setting
Figure 10-12 Crystal Trim Test Status
Figure 10-13 Frequency Test Setting
Figure 10-14 Frequency Test Status
Figure 10-15 TX Output Power Setting
Figure 10-16 TX Output Power Status
Figure 10-17 RX Sensitivity Setting
Figure 10-18 RX Sensitivity Status
Figure 10-19 Function Test Setting
Figure 10-20 Function Test Status
Figure 10-21 User Firmware/Tags Setting
Figure 10-22 Example of 500 Entries
Figure 10-23 Out of Address Error
Figure 10-24 OTP settings
Figure 10-25 Hardware EFuse settings
Figure 10-26 UART0 Pin Modification
Figure 10-27 UART Setting for EVK (EVB) and Modules

CONFIDENTIAL

Figure 10-28 Start Option
Figure 10-29 SN Text Box
Figure 10-30 MP Region Information
Figure 10-31 Write Calibration Data to MP Region
Figure 10-32 Command Mode Setting
Figure 10-33 Command Menu
Figure 10-34 PV Test Setup
Figure 10-35 PV Test Setting
Figure 10-36 Get custom NVDS with Call External Process
Figure 10-37 Update Test Result with Call External Process
Figure 11-1 CLI Command Detail Logs
Figure 16-1 No Board Found
Figure 16-2 Different Type of Boards Found Simultaneously Error Log
Figure 16-3 Invalid Board <num> Found

List of Tables

Table 2-1 APT and DL Kit Information
Table 2-2 Supported Atmosic Devices
Table 14-1 nvds_tool Options List
Table 15-1 NVDS/MP Region and OTP Tag ID

CONFIDENTIAL

Acronyms and Abbreviations

Acronyms	Definition
APT	Atmosic Production Tester
ATM2	ATM2201 ATM2202 ATM2221 ATM2231 ATM2251
ATM3	ATM3201 ATM3202 ATM3221 ATM3231
ATM33/e	ATM3325 ATM3330 ATM3330e
DL	Download Board
DUT	Device Under Test
EVB	Evaluation Board
EVK	Evaluation Kit
MP	Mass Production
MPTool	Mass Production Tool
NVDS	Non-Volatile Data Storage
NVM	Non-Volatile Memory
OTP	One Time Programmable
PV	Photovoltaics
RAM	Random Access Memory
RRAM	Resistive Random Access Memory
SoC	System-on-Chip

CONFIDENTIAL

1. Overview

This document describes the mass production environment to test ATM2/ATM3 and ATM33/e devices on production lines without RF testing equipment. It consists of both hardware and software items:

- MPTool is an application software (AtmMpTool.exe) running on MS Windows.
- APT provides the capabilities of Bluetooth LE testing with DUT controlled by AtmMpTool.exe. Refer to APT and DL Kit Information regarding the APT and DL. See Figure 2-1 for content of the APT Kit.
- Atmosic DL provides the capabilities of firmware, NVDS, and OTP programming by AtmMpTool.exe. only. This solution provides a cost-effective way to program the DUT.

The Atmosic MPTool software with the APT provide these functionalities:

- RF Tx Test
- RF Rx Test
- RF Frequency Test
- Crystal Trim
- Write firmware and NVDS
- PV Test

CONFIDENTIAL

2. Hardware and Software Requirements

2.1 Supported Hardware

Name	Description	Kit Part Number
Atmosic Production Tester (APT)	Production Test Board	ATMAPT-Mx2xx
Download Board (DL)	Product Firmware Download Board	ATMFDL-Mx2xx

Table 2-1 APT and DL Kit Information

ATM2	ATM3	ATM33/e

ATM2201-x0x	ATM2201-x1x	ATM3201-x0x	ATM3201-x1x	ATM3330e-5DCAQN
ATM2202-x0x	ATM2202-x1x	ATM3202-x0x	ATM3202-x1x	ATM3330-5DCAQN
ATM2221-x0x	ATM2221-x1x	ATM3221-x0x	ATM3221-x1x	
	ATM2231-x1x		ATM3231-x1x	
	ATM2251-x1x			

Table 2-2 Supported Atmosic Devices



CONFIDENTIAL

Figure 2-1 Atmosic Production Tester Kit

2.2 Supported Software

- MPTool software (version 2.1.8.1 or later)

2.3 Supported OS

- Windows 10, Windows 11

3. Environment Setup for the APT

A complete set of the testing suite requires:

- PC x 1 (Windows 10, Windows 11)
- APT
- Dipole antenna[1] x 1
- Type-B USB cable x 1
- 2x10 Test interface cable x 1 (connect from DUT to J3 of APT)
- MPTool software

Figure 3-1 depicts the APT board configuration.

CONFIDENTIAL

[1] The dipole antenna should be connected to SMA_DIR by default.

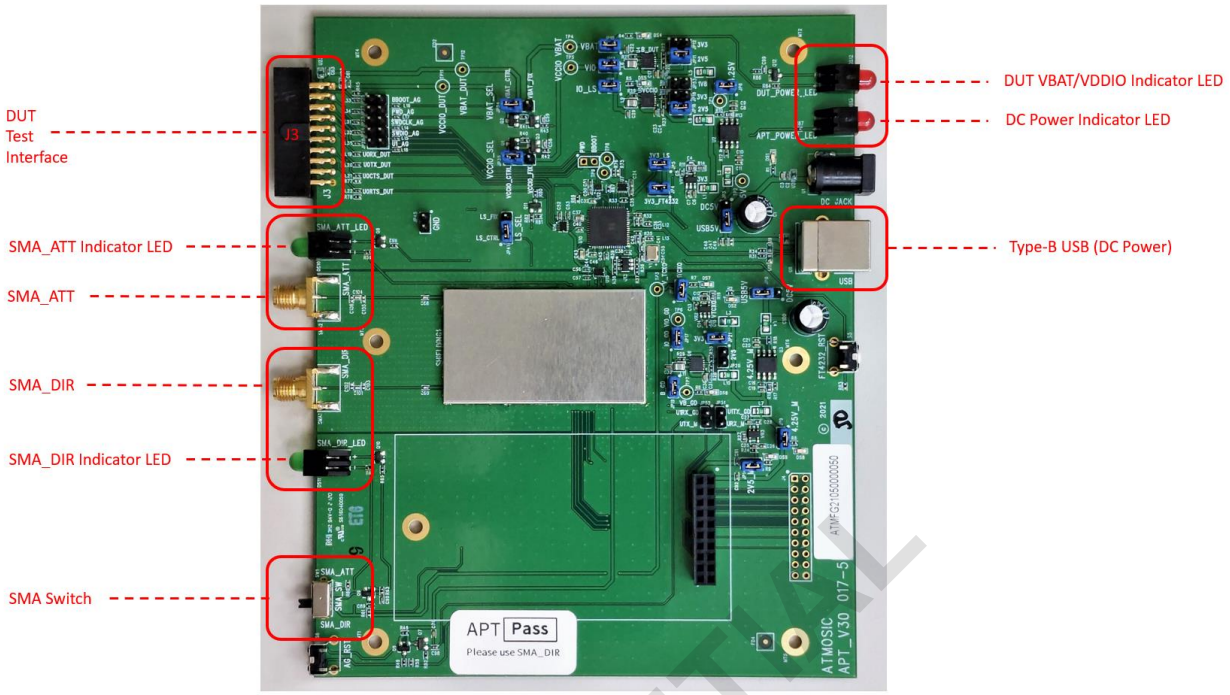


Figure 3-1 Atmosic Production Tester Board Configuration

Figure 3-2 shows a connection setup between PC, APT, and DUT:

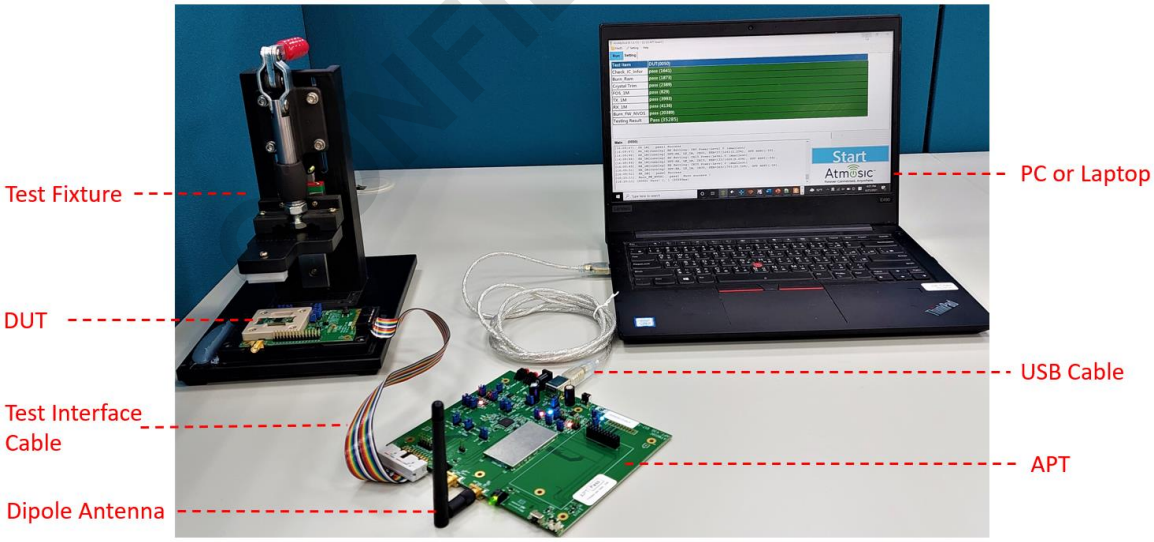


Figure 3-2 Connection Setup Between PC, APT, and DUT

CONFIDENTIAL

4. Environment Setup for Download (DL) Board

A complete set of the testing suite requires:

- PC x 1 (Windows 10, Windows 11)
- Atmosic DL Board (provided by Atmosic)
- Type-B USB cable x 1
- 2x10 Test interface cable x 1 (connect from DUT to J1 of DL)
- MPTool software

The DL_V6 picture is shown in Figure 4-1.

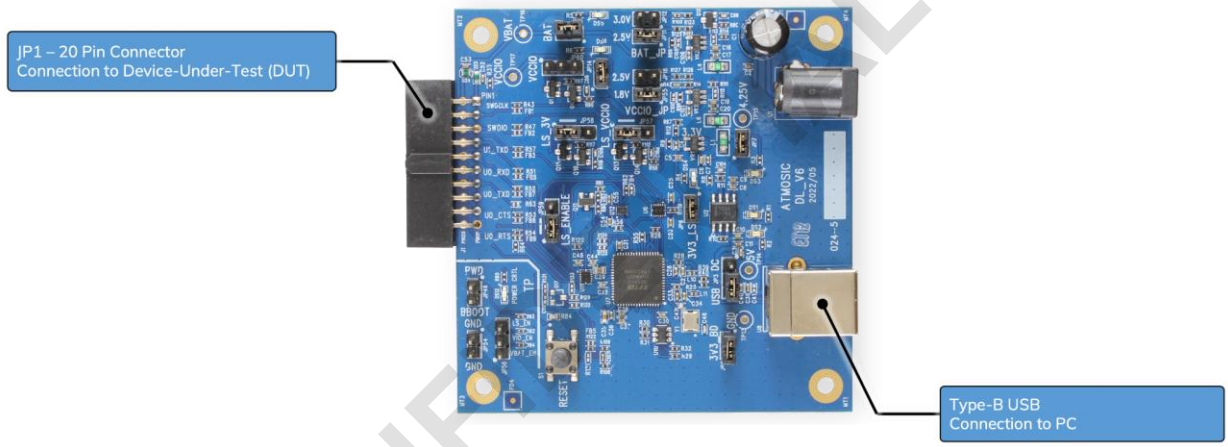


Figure 4-1 Production Download Board (DL) Configuration

CONFIDENTIAL

Figure 4-2 shows a connection setup between PC, DL, and DUT.

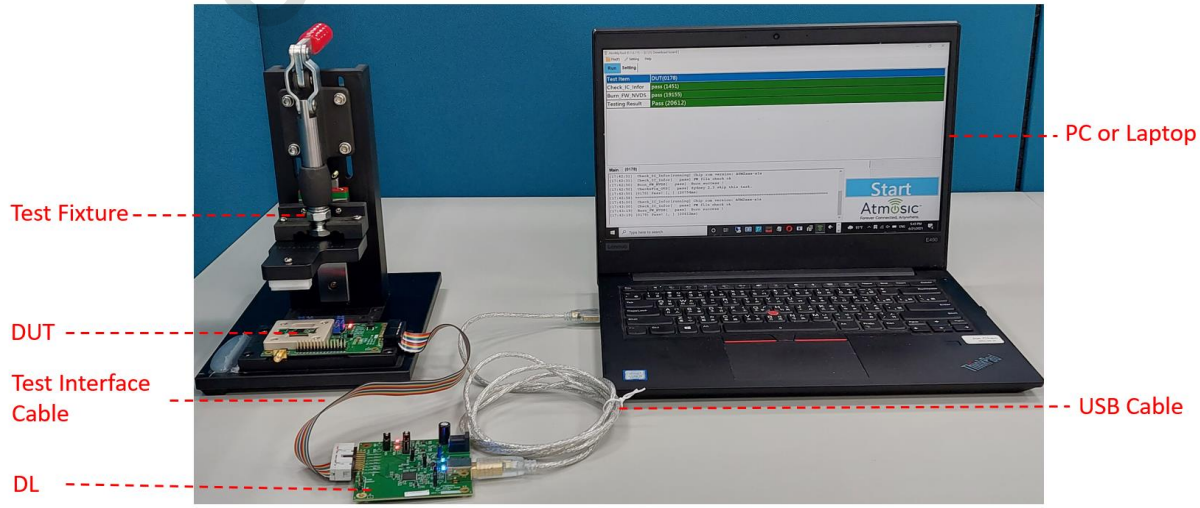


Figure 4-2 Connection Setup Between PC, DL, and DUT

4.1 DL Board Versions

4.1.1 DL_V6a

DL_V6a does not support the power control function available in DL_V6. The default jumper settings for DL_V6a are shown in Figure 4-3.

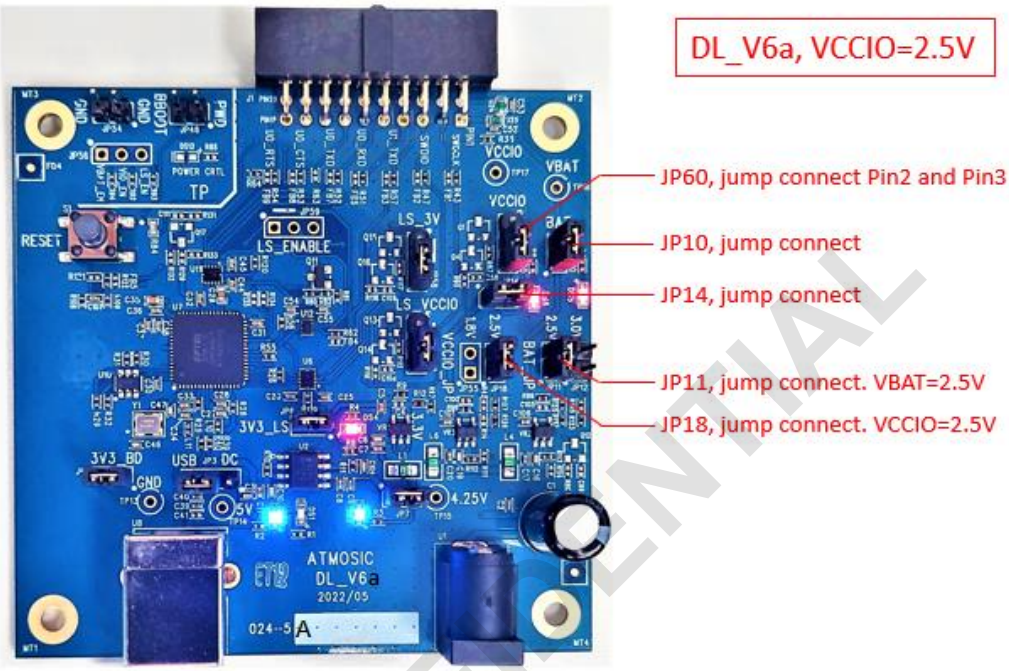


Figure 4-3 DL_V6a Critical Jumper Setting

4.1.2 DL_V6

DL_V6 adds a power control function to ensure that the DUT is powered off before the MPtool test program runs and DL_V6 supports either 2.5 V or 1.8 V I/O voltages. To use DL_V6, the MPTool version must be V2.0.12 or higher.

Since the I/O voltage needed to write into OTP requires a 2.5 V I/O voltage level, the default jumper settings are shown in Figure 4-4, which support a 2.5 V I/O voltage.

CONFIDENTIAL

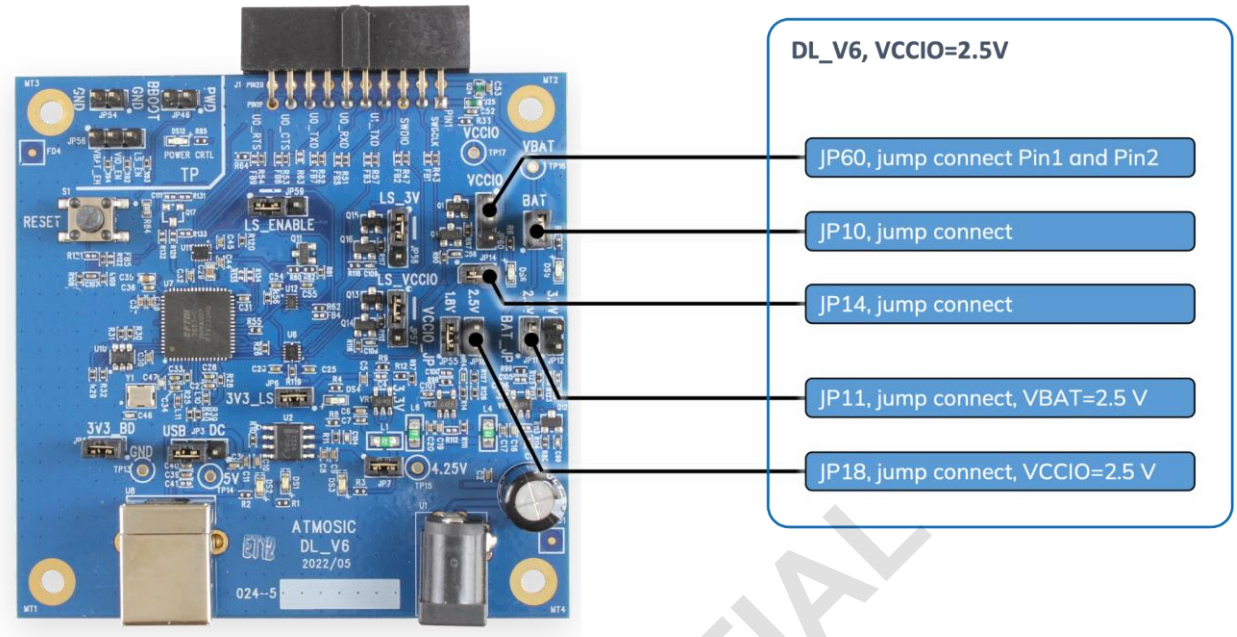


Figure 4-4 DL_V6 with 2.5 V I/O Voltage Critical Jumper Setting

If the I/O voltage needs to use 1.8 V I/O, the recommended jumper settings diagram is shown in Figure 4-5, which support 1.8 V I/O.

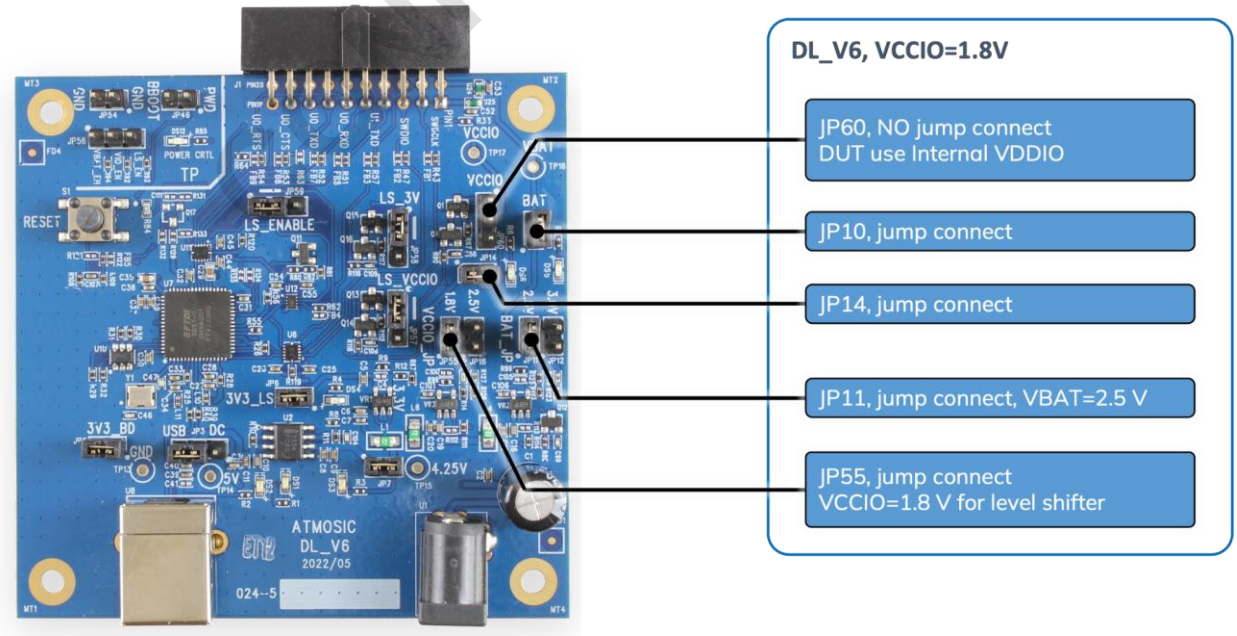


Figure 4-5 DL_V6 with 1.8 V I/O Voltage Critical Jumper Setting

CONFIDENTIAL

5. Hardware Setup

There is a Type-B USB port on the APT/DL board that is connected to an FTDI FT4232H chip, which provides four UART ports.

Test point connector is the interface between the DUT and APT/DL as shown in Figure 5-1. It provides a host computer access to some commonly needed interfaces on an Atmosic device based DUT, such as:

- Power and Ground:
  - VBAT
  - VCCIO
  - GND
- Serial Wire Debug (SWD):
  - SWDCLK
  - SWDIO
- 4-wire HCI UART for RF Test control:
  - UART0_TXD
  - UART0_RXD
  - UART0_CTS
  - UART0_RTS
- Status Control signal:
  - PWD: Used to Reset/PWD the DUT.
  - BBOOT: Used to configure the DUT into CPU idle state for programming OTP and flash.
  - TMC: Test Mode Control, pull high to enable Test Mode, which enables SWD. If TMC is already pulled high on the DUT side, this pin doesn't need to be connected.
- Debug UART:
  - UART1_TXD: Optional, for debugging only.

CONFIDENTIAL

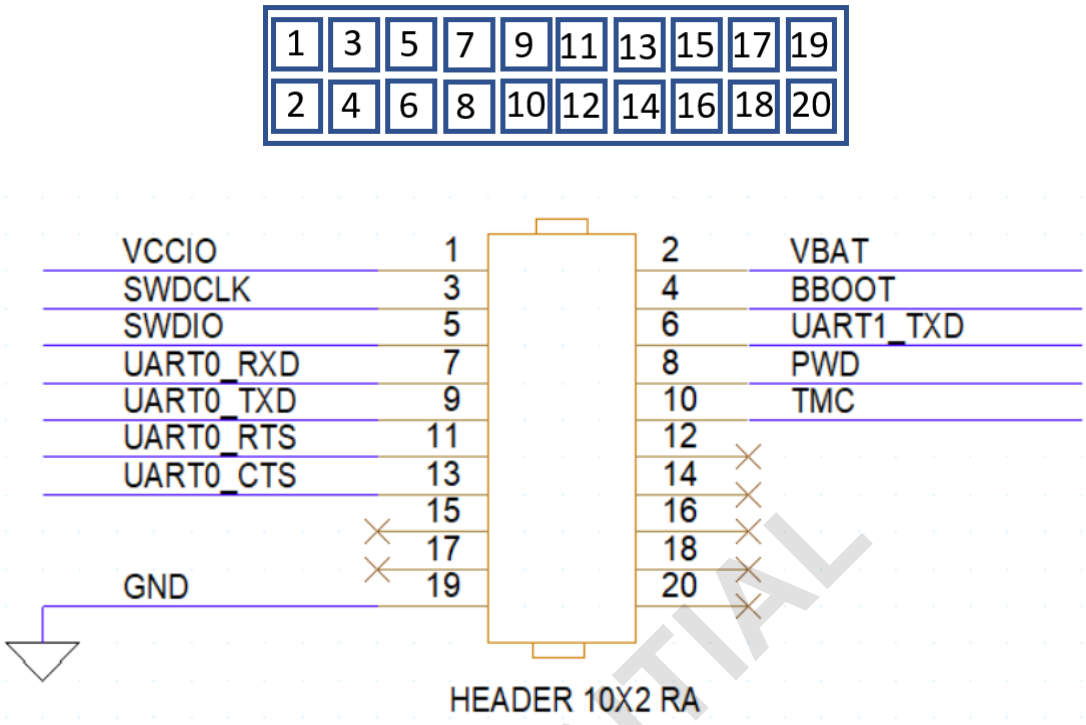


Figure 5-1 Test Point Connector Pin Definition

CONFIDENTIAL

6. DUT Design Requirements for Manufacturing

The DUT must have a matching set of headers or test points that allow the APT/DL board to connect to it for downloading firmware or executing mass production tests by the MPTool. The interface between the DUT and APT/DL is shown in Figure 5-1.

The MPTool's configuration for UART to GPIO mapping must match the DUT. This configuration is detailed in UART0 Pin Modification section.

7. Software Setup

7.1 Uninstall Any Previous Versions of the MPTool

If any earlier version of the MPTool was installed, execute its uninstaller, which should be at C:\AtmosicMP\unInst_MP.exe and wait for the uninstaller to complete, then click Close. See Figure 7-1.

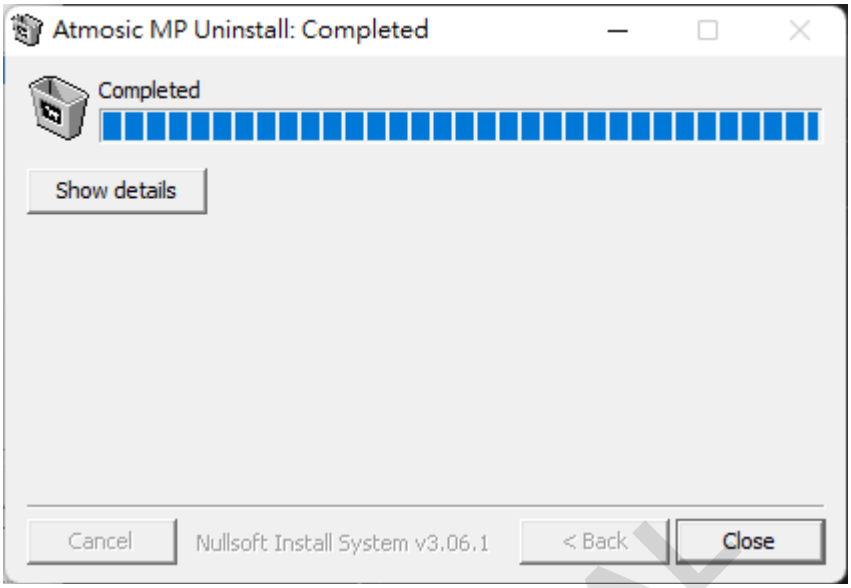


Figure 7-1 Uninstall Previous MPTool Version Completed

7.2 Install Atmosic MPTool

The setup package of Windows Installer for the Atmosic MPTool is a compressed file named **AtmosicMP_Inst_20xxxxxx_vx.x.x.x.zip**. Please extract the file and double click the AtmosicMP_Inst_20xxxxxx_vx.x.x.x.exe to start the Atmosic MPTool Installer.

The Atmosic MPTool will occupy up to 75 MB of disk space and should be located in the C:\AtmosicMP folder. Please ensure enough disk space is available before installing the Atmosic MPTool. Subsequently, click the `Install` button to continue. Figure 7-2 shows the Atmosic MPTool Destination Folder.

CONFIDENTIAL

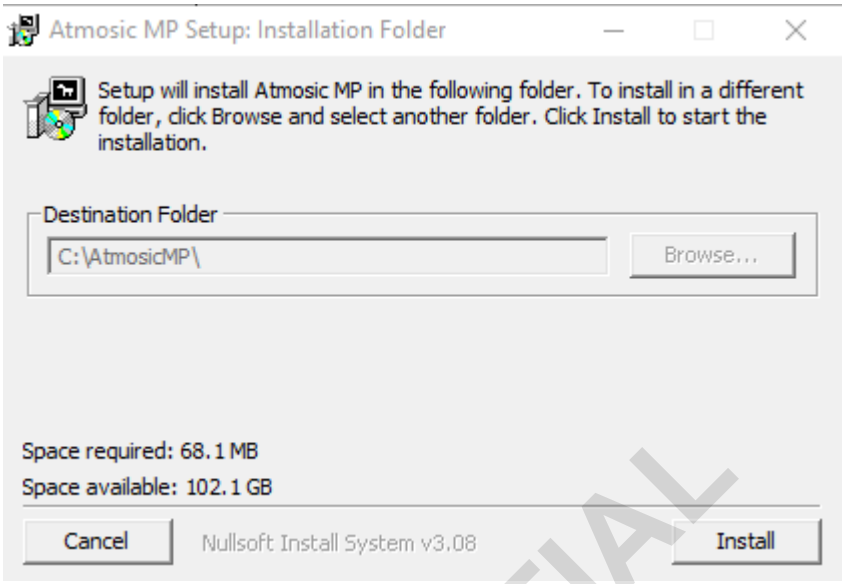


Figure 7-2 Atmosic MPTool Destination Folder

Figure 7-3 shows the screen after the installer is completed. The installer will create a folder named AtmosicMP in Start Menu\Atmosic, and create shortcuts to invoke Atmosic MPTool features that will be discussed in the next section.

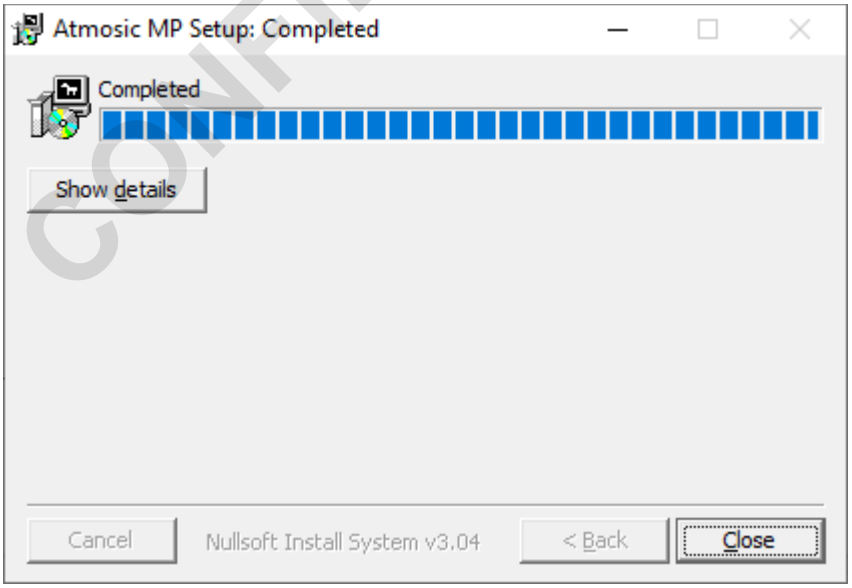


Figure 7-3 Atmosic MPTool Setup Completed

CONFIDENTIAL

7.3 Atmosic Production Tool Shortcuts

The Atmosic Production Tool Installer will create shortcuts in Start Menu\AtmosicMP after the installer is completed. The shortcuts include install or remove MFx interface and MP API document as shown in Figure 7-4.



Figure 7-4 Start Menu Shortcuts

7.4 Install MFx Interface

Please connect the Atmosic APT to the laptop and wait a while (~ 30 seconds) to install the FTDI driver, which usually exists in the Windows OS. If your computer cannot install this automatically, please download D2xx driver from FTDI website and install it yourself.

Next, open the Windows Device Manager (open the Run dialog box by pressing and holding the Windows key, then press the R key).

Next, type in devmgmt.msc then click the OK button to verify whether the Atmosic MFx interface existed or not.

If there are 4 COM ports listed in the Device Manager after plug-in of the Atmosic APT to the laptop as shown in Figure 7-5, please click the Install MFx Interface shortcut in Start Menu/AtmosicMP folder. The COM port numbers were assigned by the Windows OS.

If not, please download the FTDI driver and install it before installing the MFx interface driver.

CONFIDENTIAL

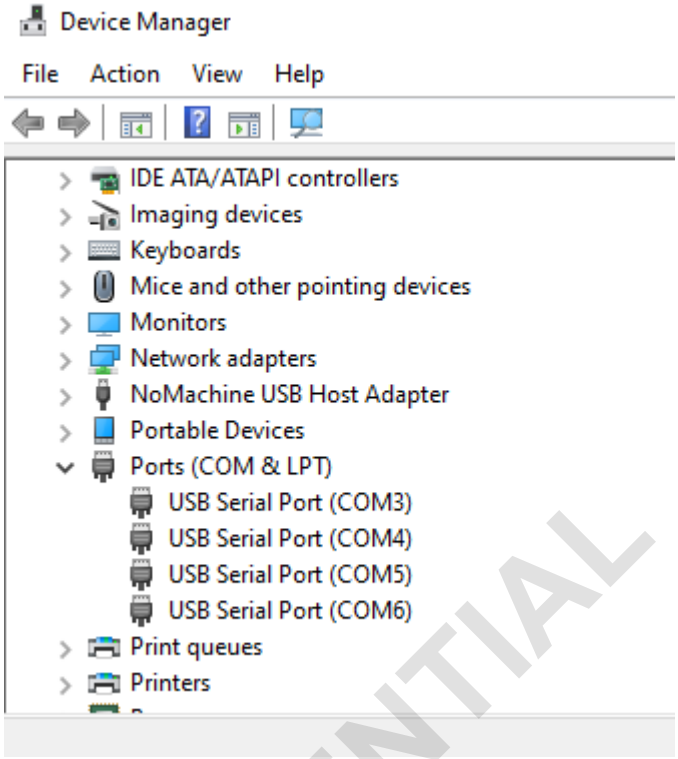


Figure 7-5 COM Port Setup in Device Manager

CONFIDENTIAL

- The first COM port (such as COM3 in Figure 7-5) is used to program firmware, NVDS, or OTP. (Need to change to Atmosic MFx driver)
- The second COM port (such as COM4 in Figure 7-5) is used to dump logs from DUT.
- The third COM port (such as COM5 in Figure 7-5) is used to communicate between Atmosic DUT and laptop through HCI commands.
- The fourth COM port (such as COM6 in Figure Figure 7-5) is used to communicate between Atmosic APT and laptop through HCI commands.

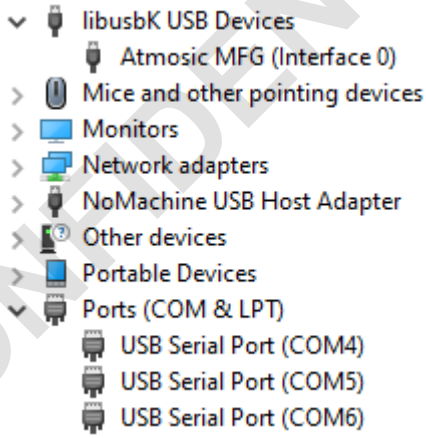
Next, install the Atmosic MFx driver, see Figure 7-6.

You can find the install under the Start Menu/AtmosicMP folder `Install MFx Interface`.



Figure 7-6 Atmosic MFx Driver

After installing the Atmosic MFx driver, the first COM port will change to libusbK USB Devices (such as Atmosic MFG (Interface 0) in Figure 7-7).



CONFIDENTIAL

Figure 7-7 COM Port Assignment

7.5 Uninstall MFx Interface

All Atmosic MFx Interfaces that are installed in the Windows OS will be uninstalled after performing the Uninstall MFx Interface program.

8. Adjust Serial Port Latency Timer

Setting a lower latency timer will speed up the testing process when the test item uses the UART HCI command.

First, open the Windows Device Manager (open the Run dialog box by pressing and holding the Windows key, then press the R key).

Next, type-in devmgmt.msc then click the OK button to verify whether the Atmosic MFx interface existed or not.

As shown in Figure 8-1, for every USB Serial Port, please right click and select Properties.

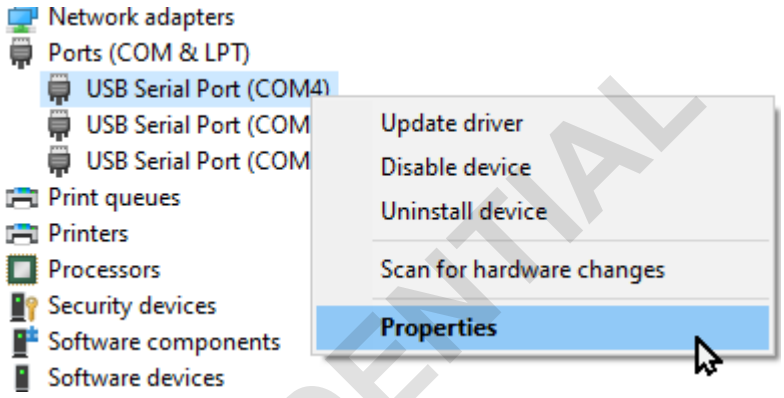


Figure 8-1 COM Port Setup in Device Manager

See Figure 8-2, select Port Settings, then click Advanced… button.

CONFIDENTIAL

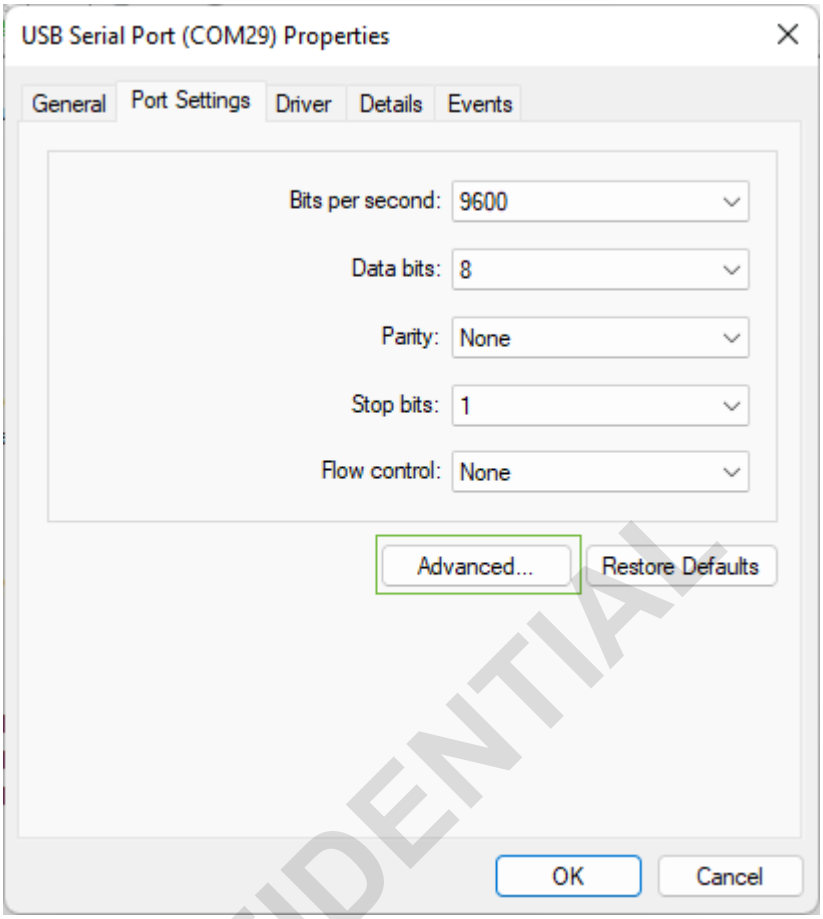


Figure 8-2 USB Serial Port Properties

CONFIDENTIAL

See Figure 8-3, change `Latency Timer` to 1 msec (millisecond)

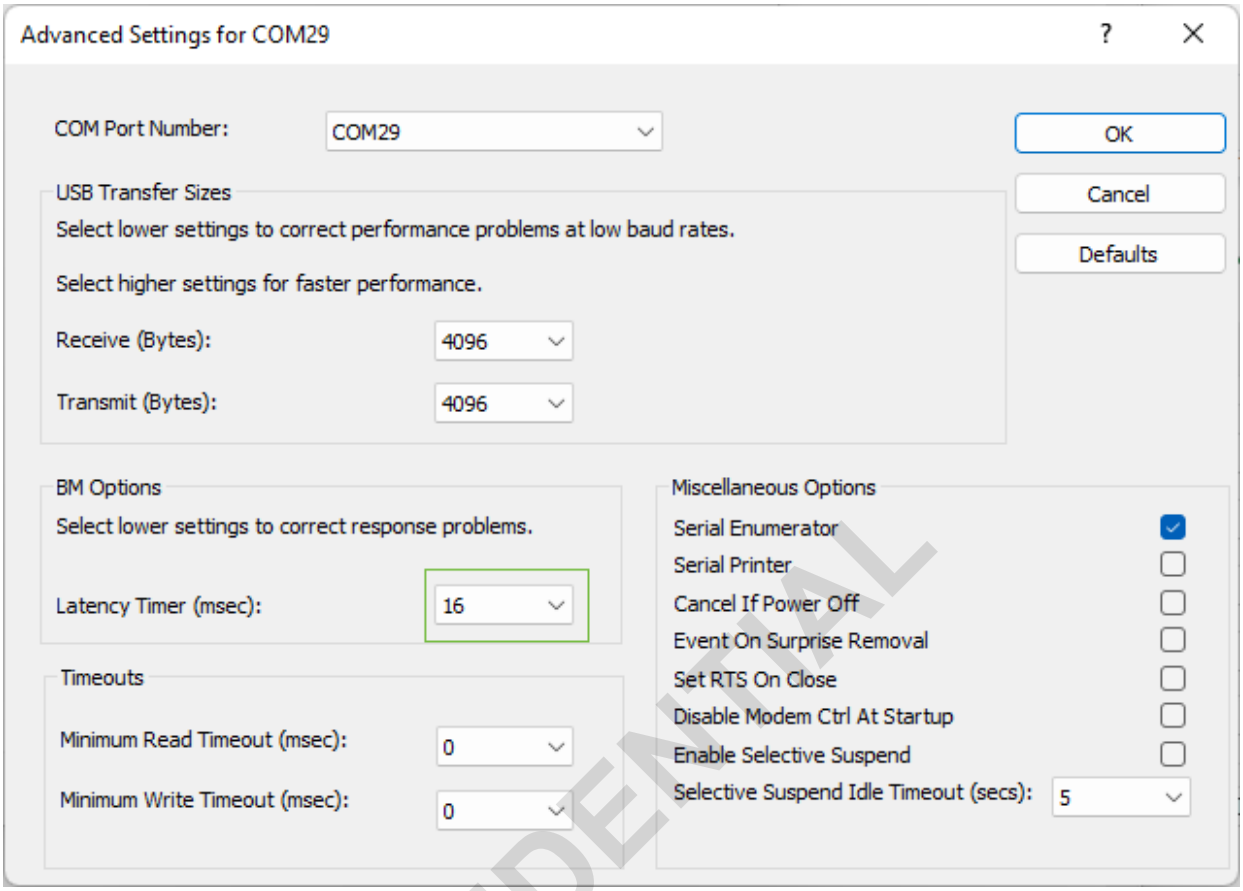


Figure 8-3 Advanced Setting for COM ports

CONFIDENTIAL

9. GUI Mode Quick Start

In GUI mode, please refer to the settings in this chapter first, following sections are steps to quickly start testing. Launch the GUI MPTool from Start Menu/AtmosicMP folder.

9.1 Select ATM File

If you want to program flash, please choose an ATM file as shown in Figure 9-1.

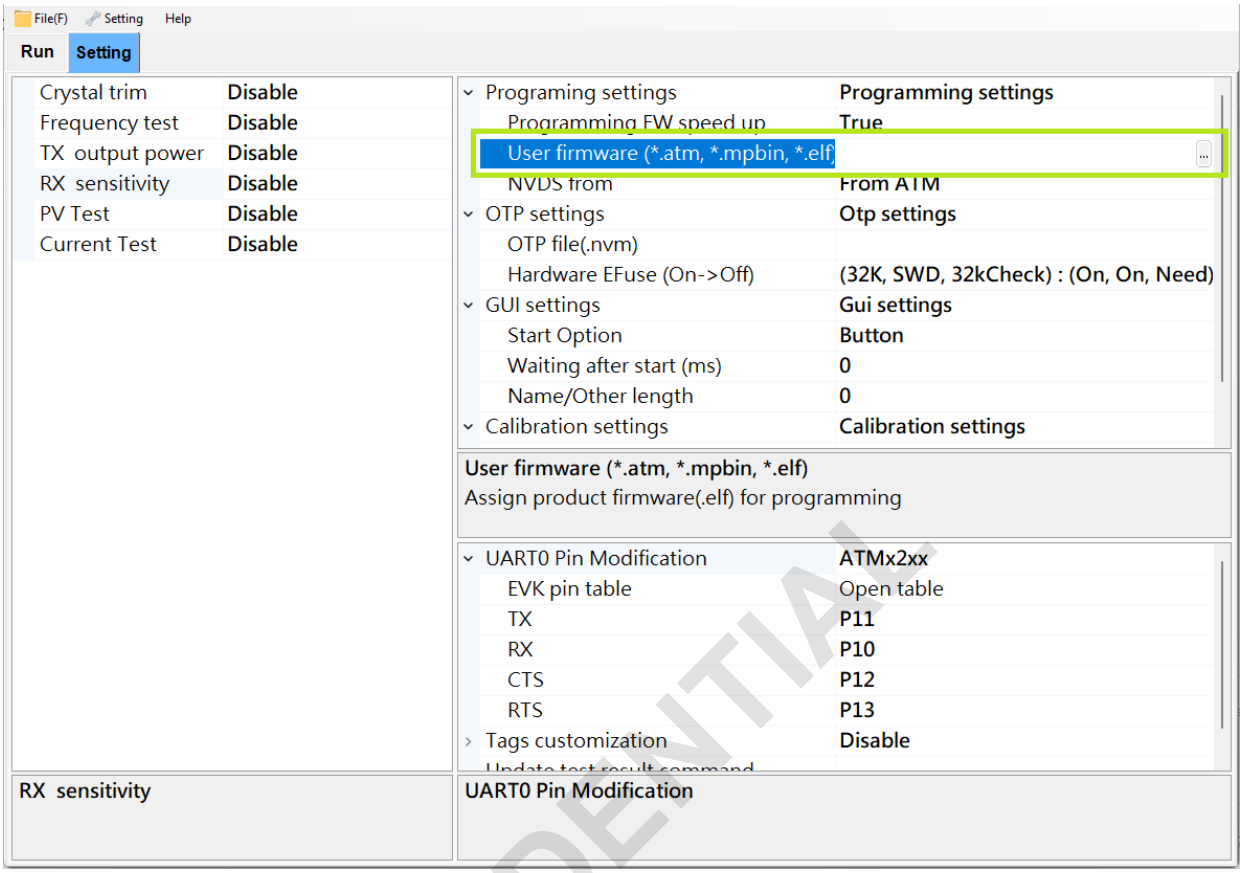


Figure 9-1 Select ATM File

***Note**: If ATM File is not selectable, the `Setting` Tab is in a lock state. Please review [Quick Start Operation Steps] to unlock the `Setting` Tab.*

9.2 RF Testing with APT

9.2.1 Test Items

There are four test items:

- Crystal trim
- Frequency test
- TX output power
- RX sensitivity

You can change these items to `Enable` to enable these items. See Figure 9-2.

CONFIDENTIAL

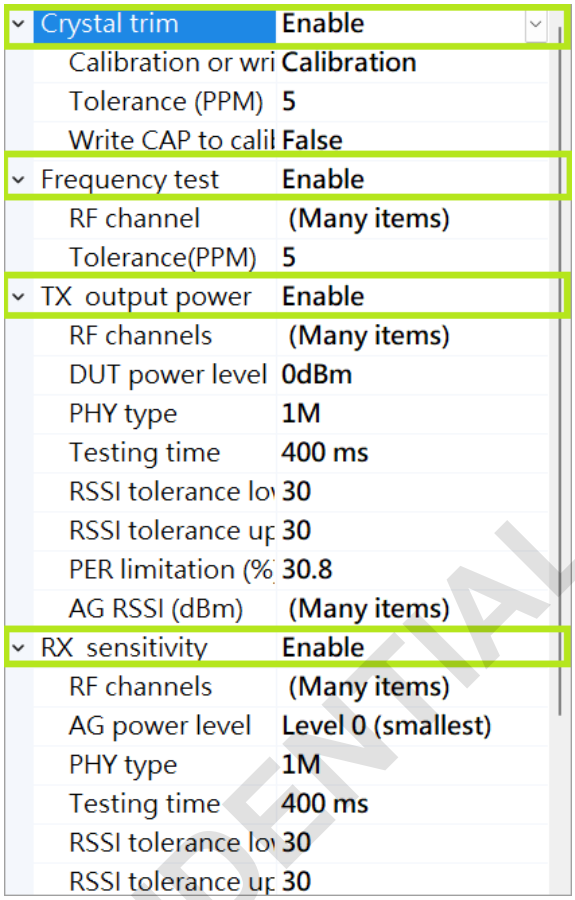


Figure 9-2 Choose Test Options

When you select any RF test, it will trigger the Burn Test Code process and it will use the HCI command via UART to perform RF testing. You can see the Burn Test Code process shown on the Run page. See Figure 9-3.

CONFIDENTIAL

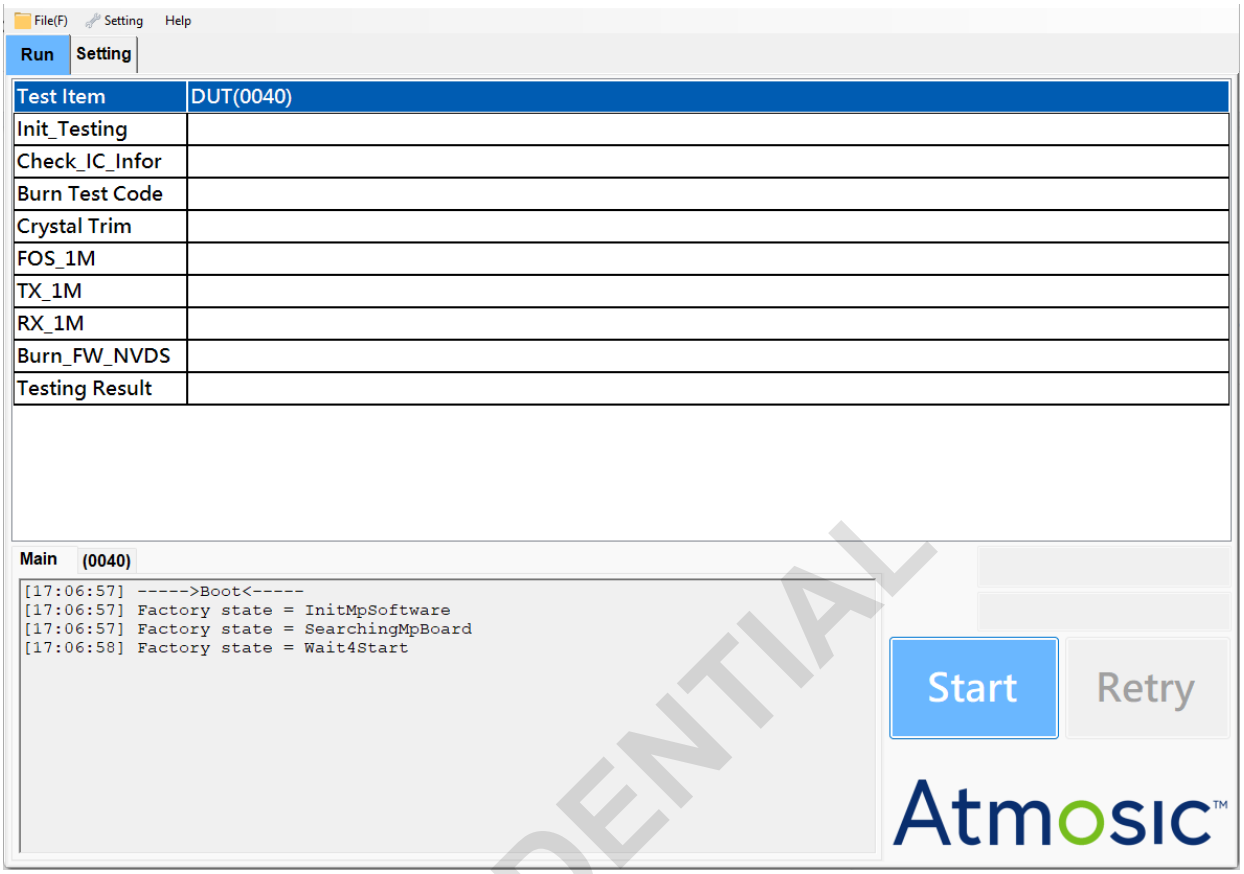


Figure 9-3 Burn_RAM Process

CONFIDENTIAL

9.2.2 Testing Options

When choosing RF testing, UART should be used to send HCI commands to the DUT, therefore, please set UART pins first.

Choose Chip Family

First, select chip family (ATMx2xx or ATM33xx):

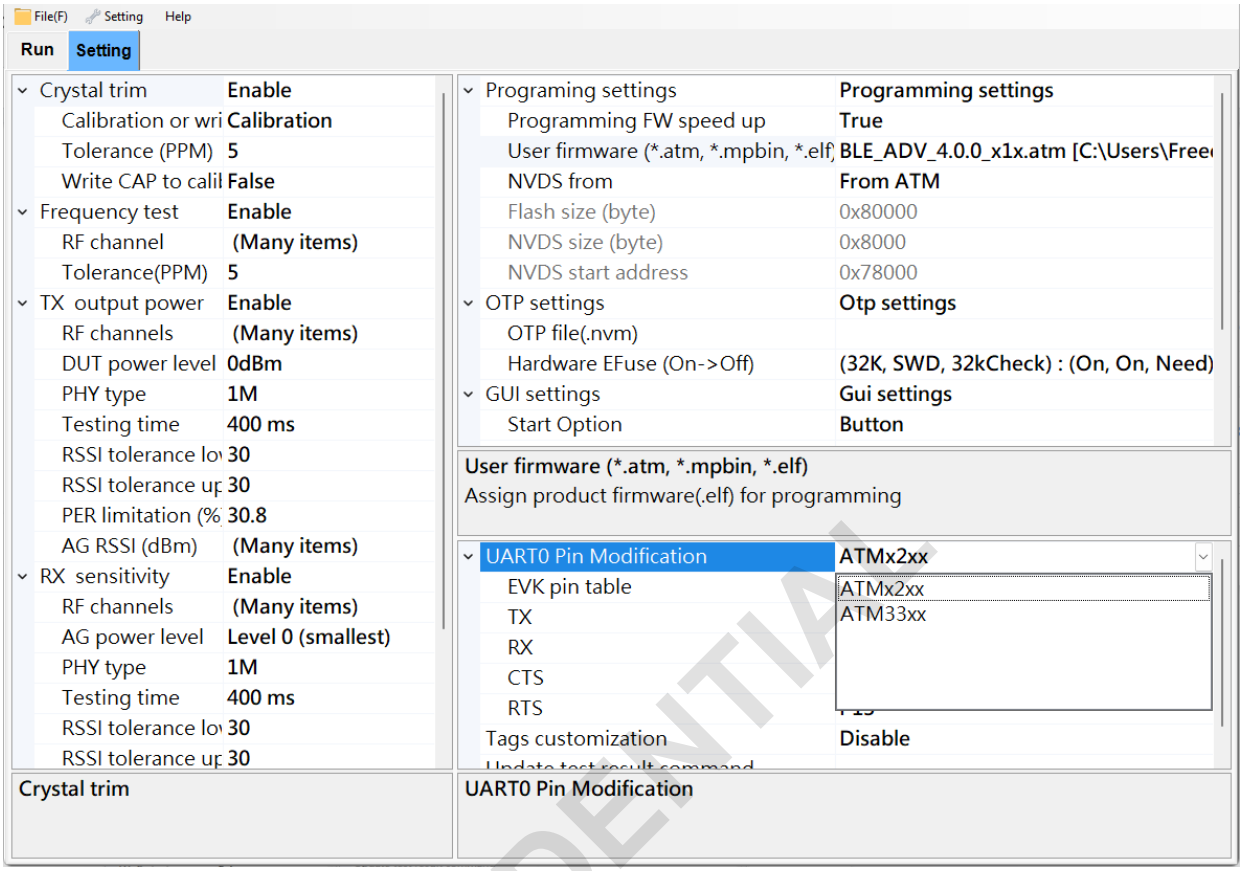


Figure 9-4 Choose Chip Family

UART0 Pin Modification

Select TX, RX, CTS, RTS as defined by your board:

CONFIDENTIAL

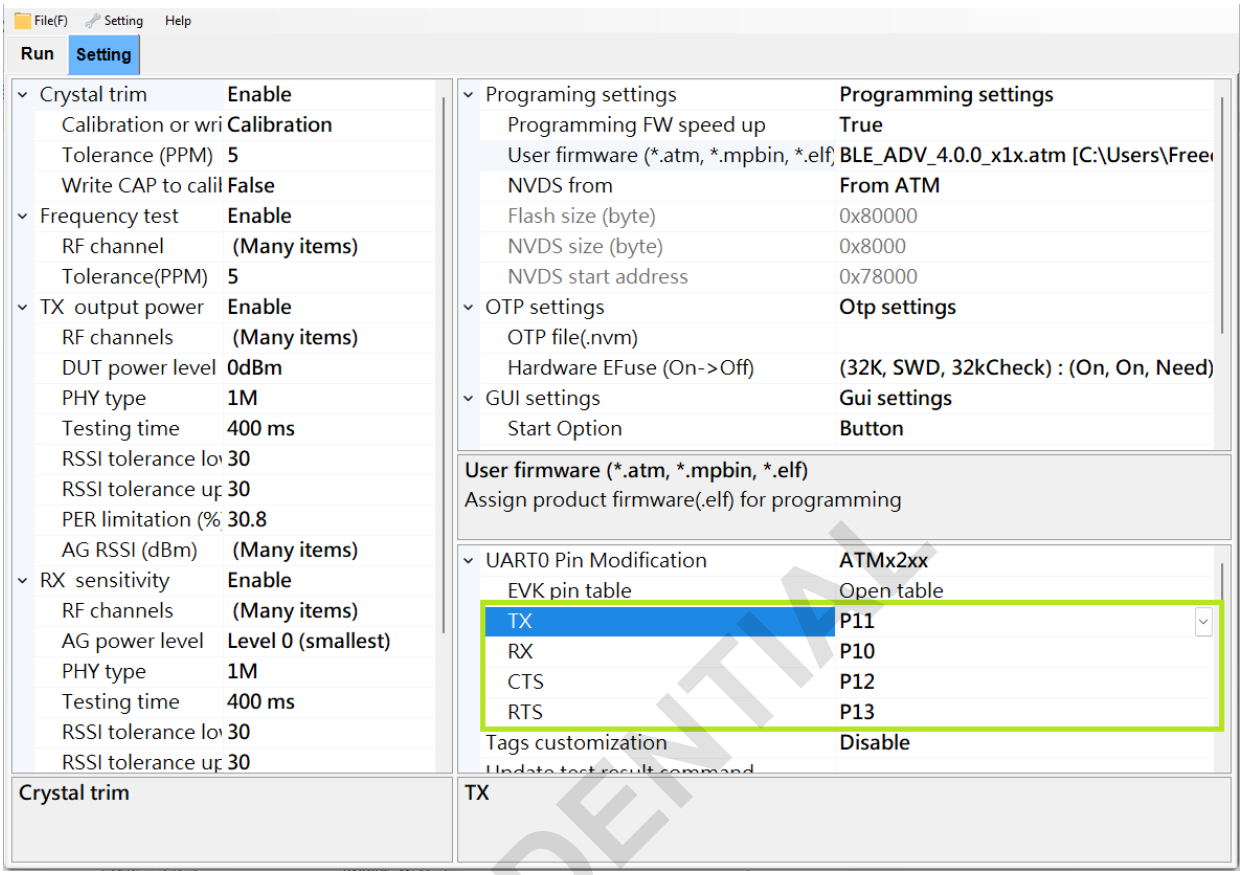


Figure 9-5 Select ATM File

CONFIDENTIAL

For more suggestions on UART pin setting, please refer to the UART0 Pin Modification section.

10. Atmosic Mass Production Tool User Interface

The MPTool is used for mass production tests on ATM2/ATM3 and ATM33/e devices. The procedure includes crystal trim, frequency test, TX test, RX test, and program firmware.

After installation, users can launch AtmMpTool.exe from the Start Menu/AtmosicMP folder of the Windows OS.

AtmMpTool.exe will generate production log files for each test. These log files are saved in the log folder. It can be opened by clicking the Open Production Log Folder menu item.

See Figure 10-1.

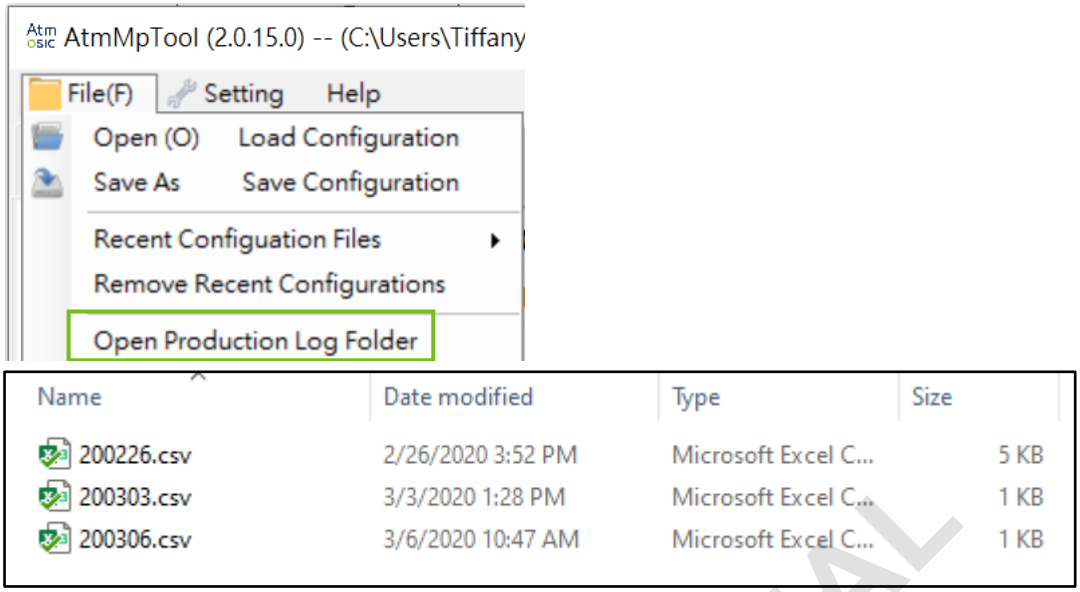


Figure 10-1 Atmosic Mass Product Tool User Interface

The production logs were written in .csv format and one file every day.

The GUI contains three parts:

- [Menus] : Used for saving config, refreshing devices, unlocking Setting Tab, and selecting language.
- [Run Tab] : Used for controlling the running of tests.
- [Setting Tab] : Used for setting the parameters of test items.

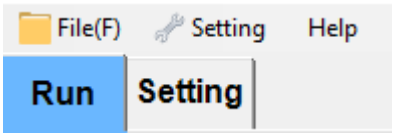


Figure 10-2 MPTool Run Menu

Quick Start Operation Steps:

1) Please unlock Setting Tab. (password is required and initial value is Atmosic123)
2) Set the test parameters of enabled test items in Setting Tab. (Or load setting from File)
3) Click the START button on the Run Tab to start the test on DUTs.

CONFIDENTIAL

10.1 Menus

10.1.1 File Menu

Here are five menu items as shown in Figure 10-3 and describe as below:

- [Open] : Load configuration file into application.
- [Save as] : Save current test configuration to file.
- [Recent Configuration Files] : Recent list of configuration files.
- [Remove Recent Configurations] : Remove the recent list.
- [Open Production Log Folder] : Open production logs' folder.

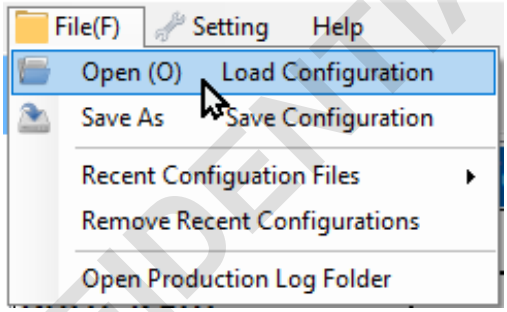


Figure 10-3 MPTool File Menu

10.1.2 Setting Menu

Here are four menu items as shown in Figure 10-4 and describe as below:

- [Unlock] : Unlock the Setting Tab from read only mode.
- [Stop] : Stop testing after testing is started.
- [Refresh APT/DL] : Re-search the APT. Generally, APTs are searched automatically during program startup. This is used when a user plugs in/out APTs without relaunching AtmMptool.exe.
- [Command mode] : Enter command line mode, user can perform the unit test by typing predefined commands.

CONFIDENTIAL

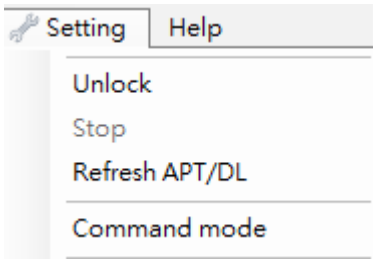


Figure 10-4 MPTool Setting

10.1.3 Help Menu

Figure 10-5 shows the MPTool Help Menu.

- [Language] : Switch the language between Simplified Chinese and English.
- [About Atmosic Manufacturing Tool] : About tool information.

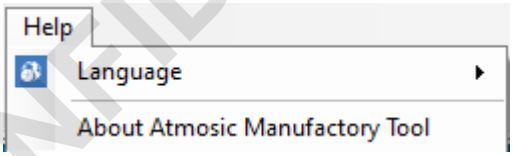


Figure 10-5 MPTool Help Menu

CONFIDENTIAL

10.2 Run Tab

There are three main parts in the Run Tab as shown in Figure 10-6:

1) Start/Stop button, used for controlling each test iteration.
2) Debug log window, show detail log of the current test.
3) Test progressing table, indicate the current testing schedule.

Run Tab before testing:

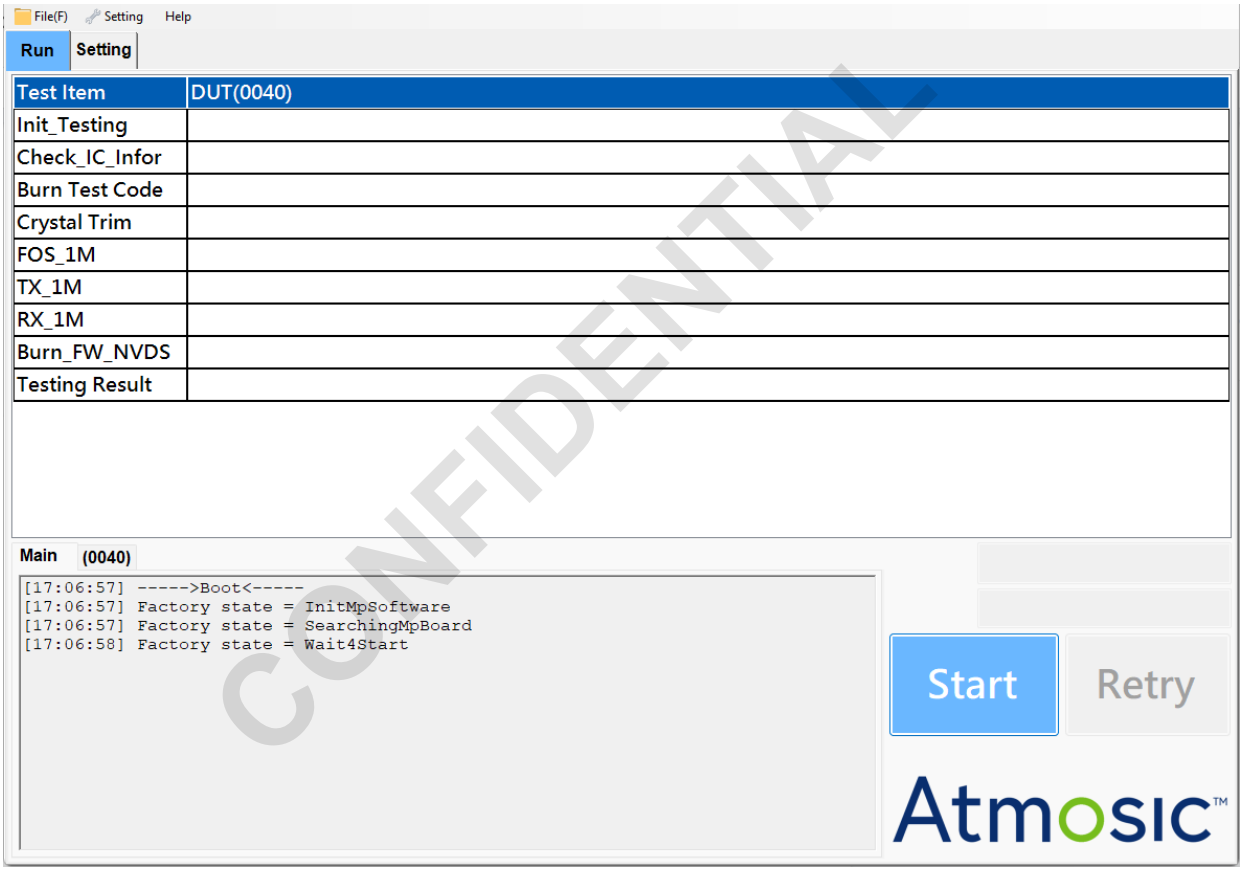


Figure 10-6 Run Tab Before Testing

Before running, all names of test items will show in the first column of the progressing table for the user to check.

Run Tab under testing:

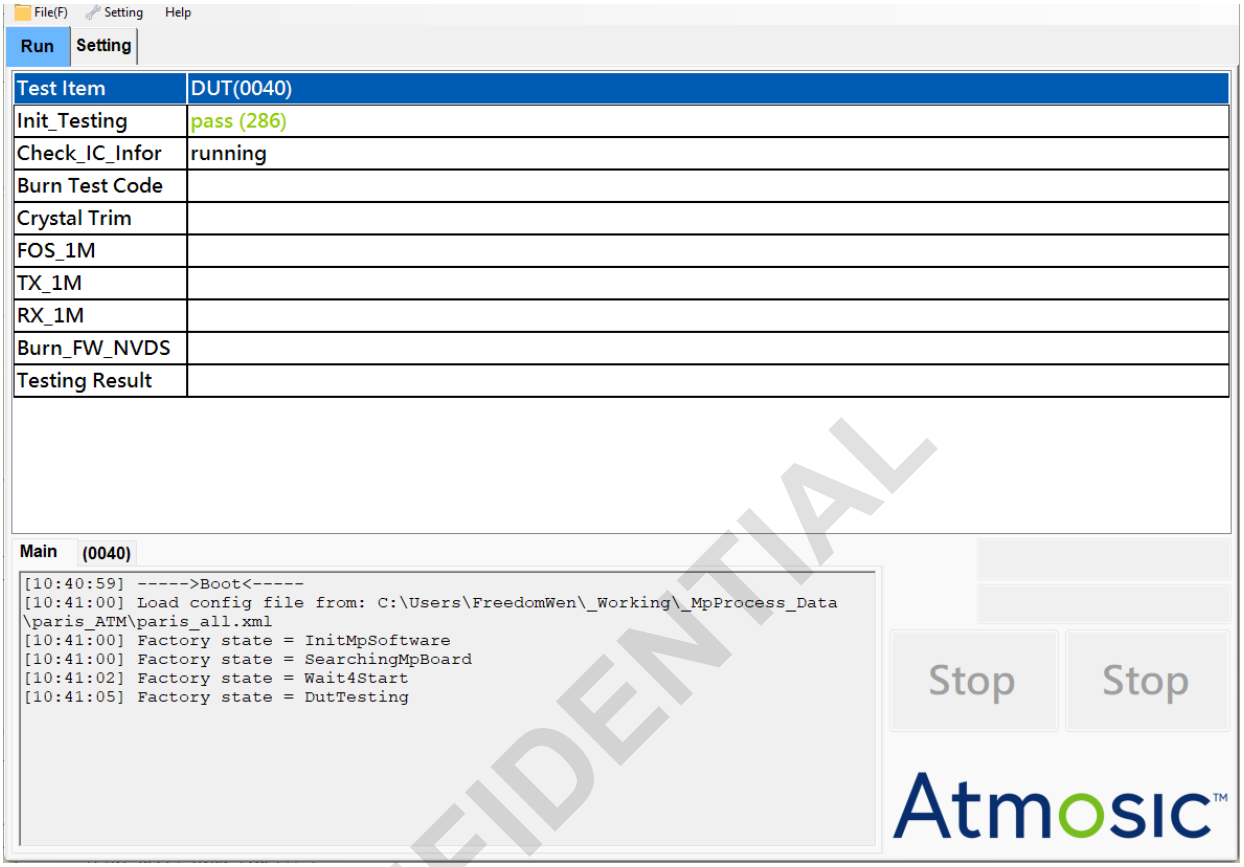


Figure 10-7 Run Tab Under Testing

When running, the corresponding row will show pass or fail and its execution time in milliseconds. See Figure 10-7.

CONFIDENTIAL

Run Tab after testing is successful:

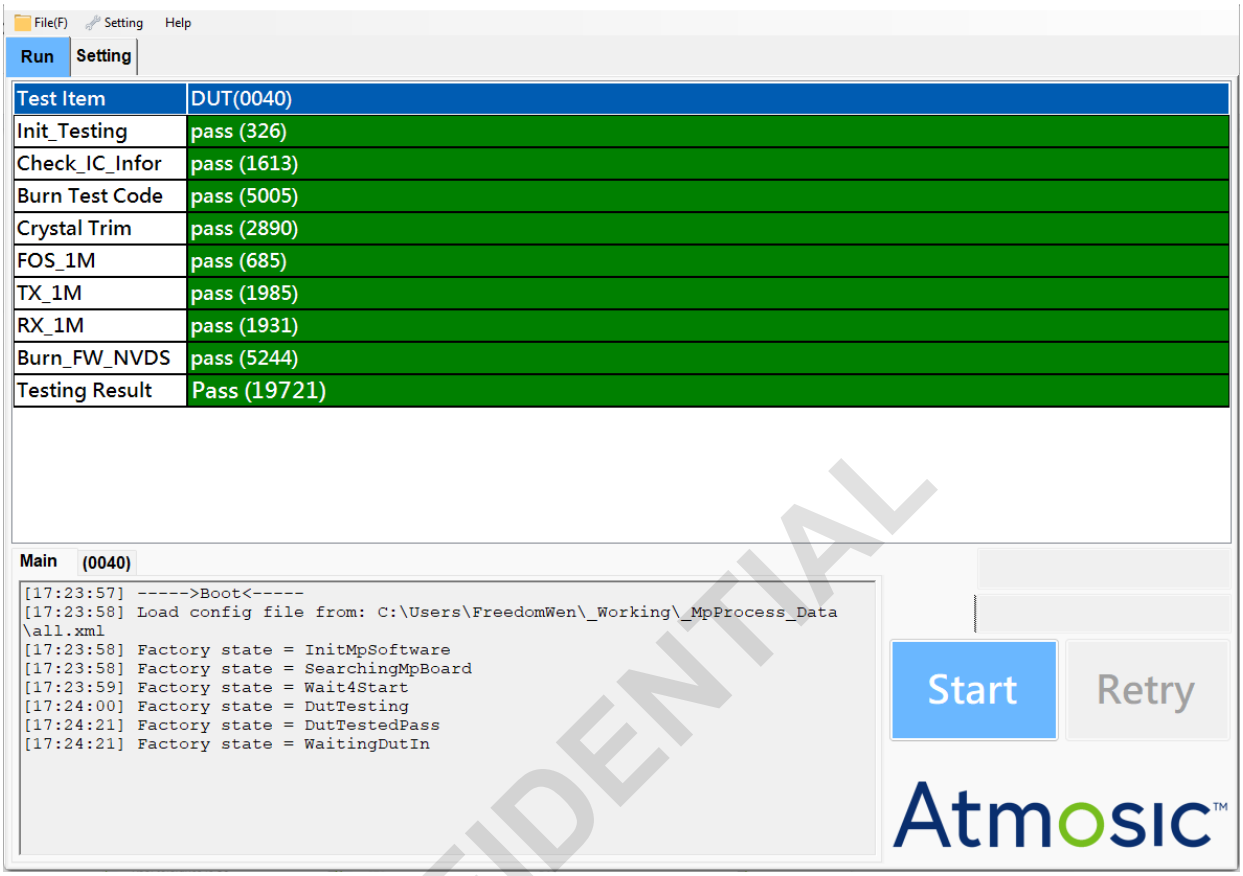


Figure 10-8 Run Tab After Testing Success

After it is finished and all passes, the background will become green. See Figure 10-8.

CONFIDENTIAL

Run Tab after testing fais:

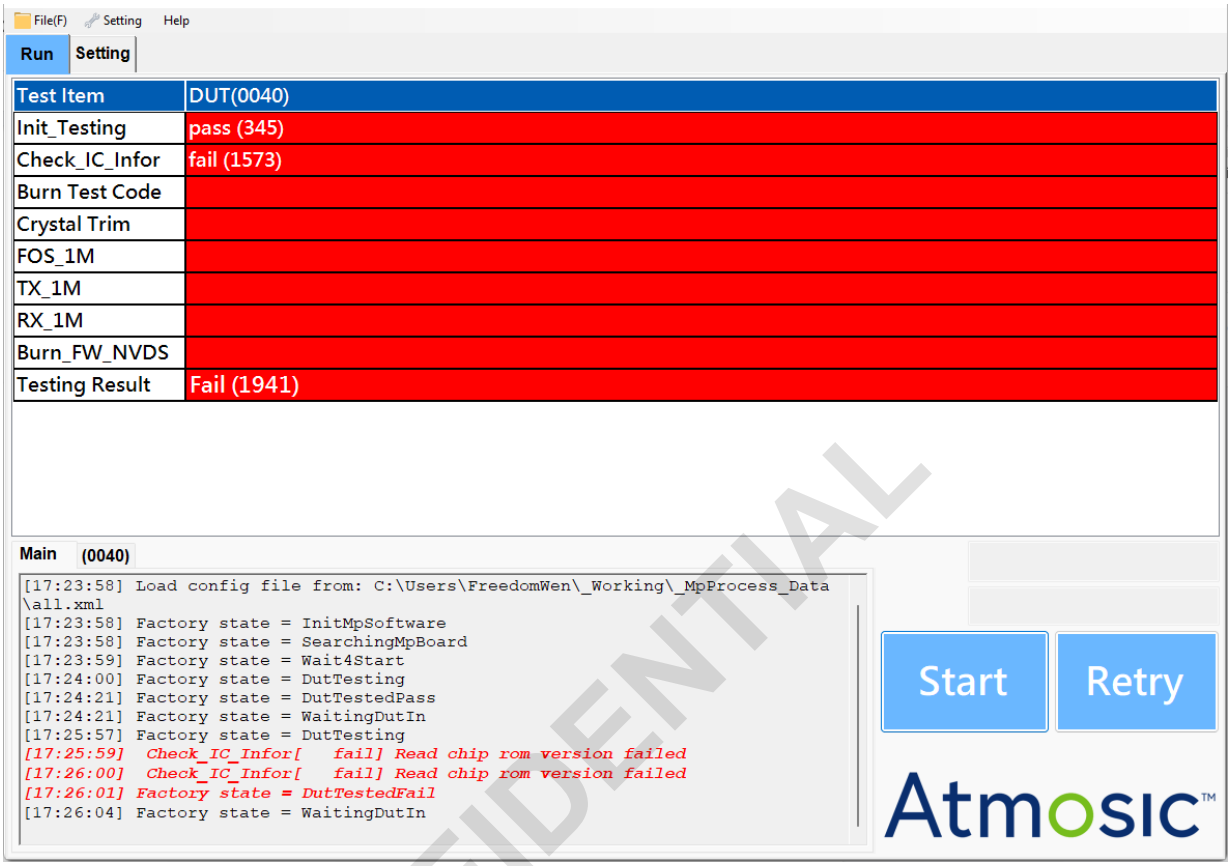


Figure 10-9 Run Tab After Testing Fails

The background color will become red if there is a failed item. See Figure 10-9.

CONFIDENTIAL

10.3 Settings Tab

There are three main parts in Setting Tab as shown in Figure 10-10:

1) Setting for Testing.
2) Setting for Programming flash, OTP, and MP region.
3) Setting for Bluetooth address and name.

To start setting the parameter, unlock the Tab by clicking on the unlock menu item.

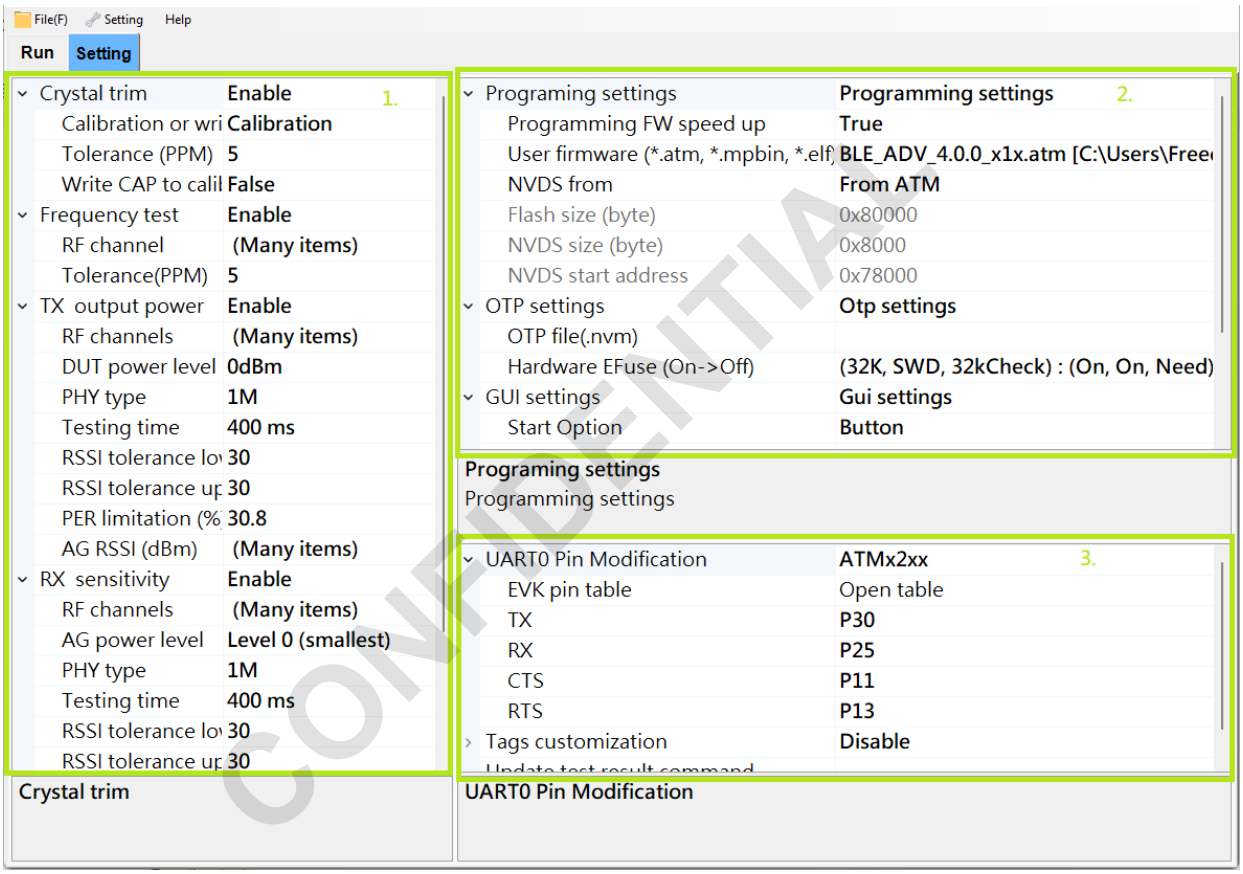


Figure 10-10 Setting Tab

For testing, there are 9 topics:

- Crystal trim
- Frequency test
- TX output power
- RX sensitivity
- User firmware(.elf) ,NVDS file (.bin),Tags customization

CONFIDENTIAL

- OTP file (.nvm)
- Write MP Region
- Photovoltaics (PV) Test

10.4 Crystal Trim

This test item trims the crystal offset by measuring the frequency offset of the crystal signal using the APT. After trimming, users can write the crystal CAP (capacitance) to OTP in ATM2/ATM3 or Secure Journal in ATM33. When performing Crystal Trim, the APT will tune the capacitance of the crystal circuit and measure the frequency offset by APT. Figure 10-11 shows the Crystal Trim Setting.

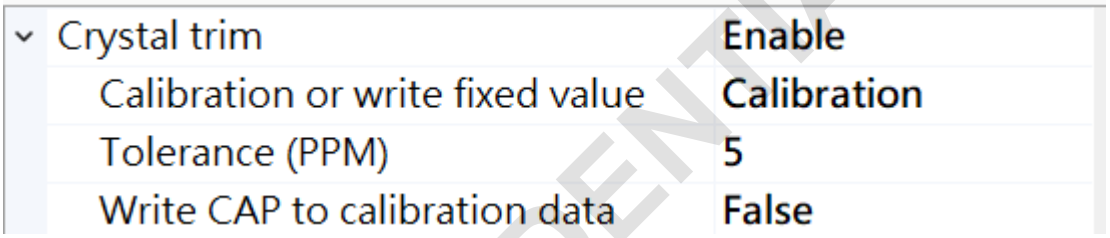


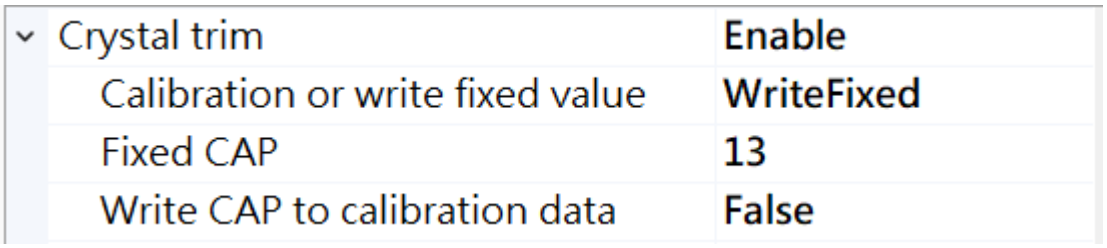
Figure 10-11 Crystal Trim Setting

CONFIDENTIAL

The Calibration or write fixed value can choose this test run calibration flow or only write fixed value.

The Tolerance (PPM) is the accuracy requirement of crystal trim.

Set the `Write CAP to calibration data` to `True` will overwrite the CAP to the calibration area.. Please see Calibration settings section.



Set the `Calibration or write fixed value` to `WriteFixed` will use the Fixed CAP value for each testing.The `Fixed CAP` is used when you want to write the same CAP for all devices.

See Figure 10-12 for test status.

```
[15:25:08]  Burn Test Code[running] started
[15:25:10]  Burn Test Code[   pass] Program RAM success.
[15:25:10]  Crystal Trim[running] started
[15:25:10]  Crystal Trim[running] 0x0D != 0x0D, trim!
[15:25:10] [Clib][Atm_MpRf16MXtalCalTest] cval: 10, Freq Delta: -22460
[15:25:11] [Clib][Atm_MpRf16MXtalCalTest] cval: 08, Freq Delta:  67626
[15:25:12] [Clib][Atm_MpRf16MXtalCalTest] cval: 0C, Freq Delta:  14159
[15:25:12] [Clib][Atm_MpRf16MXtalCalTest] cval: 0E, Freq Delta: -5126
[15:25:13] [Clib][Atm_MpRf16MXtalCalTest] cval: 0D, Freq Delta:  3906
[15:25:13] [Clib][Atm_MpRf16MXtalCalTest] trim result fval: 0D, Freq Delta:  3173
[15:25:14]  Crystal Trim[   pass] OK, CAP: 0x0D FREQ: 3173Hz
[15:25:14]  FOS_1M[running] started.
```

Figure 10-12 Crystal Trim Test Status

10.5 Frequency Test

This test measures the frequency offset of the radio. Figure 10-13 shows the Frequency Test Setting.

The `Tolerance (PPM)` is the accuracy criteria by percentage per million (PPM) for passing this test.

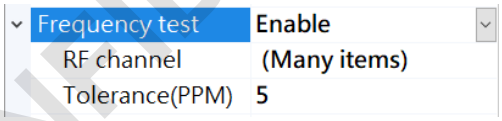


Figure 10-13 Frequency Test Setting

After starting the test, DUT will start to transmit continuous TX and APT will measure the frequency of the DUT. The DUT's frequency must be in the target frequency ± `Tolerance (PPM)`.

Figure 10-14 shows the test status.

```
[15:25:14]  Crystal Trim[   pass] OK, CAP: 0x0D FREQ: 3173Hz
[15:25:14]  FOS_1M[running] started.
[15:25:15] [Clib][Atm_MpRfFosTest] Freq Delta: 4882
[15:25:15]  FOS_1M[running] CH12: Frequency offset = 4882Hz, 2.0ppm
[15:25:15]  FOS_1M[   pass] pass
[15:25:15]  TX_1M[running] started
[15:25:15]  TX_1M[running] TX Setting: LE_1M, CH00, 0dBm
[15:25:15]  TX_1M[running] DUT: TX, LE_1M, CH00, PER=11/767(1.43%), RSSI(-44)
[15:25:15]  TX_1M[running] TX Setting: LE_1M, CH12, 0dBm
```

Figure 10-14 Frequency Test Status

CONFIDENTIAL

10.6 TX Output Power

DUT transmits continuous TX packets on the specified `RF channels`, `DUT power level` and `PHY type`, and the APT measures the DUT RSSI and calibrates DUT's output power. Figure 10-15 shows the TX Output Power Setting.

TX output power	Enable
RF channels	(Many items)
DUT power level	0dBm
PHY type	1M
Testing time	400 ms
RSSI tolerance lower (dB)	30
RSSI tolerance upper (dB)	30
PER limitation (%)	30.8
AG RSSI (dBm)	(Many items)

Figure 10-15 TX Output Power Setting

CONFIDENTIAL

The test result should be in the range of `DUT power level` ± `RSSI Tolerance (dB)`. Otherwise, it will report failure. Figure 10-16 shows the TX Output Power Status.

```
[13:01:47]  TX_1M[running] started
[13:01:47]  TX_1M[running] TX Setting: LE_1M, CH00, 0dBm
[13:01:48]  TX_1M[running] DUT: TX, LE_1M, CH00, PER=0/770(0.00%), RSSI(-57)
[13:01:48]  TX_1M[running] TX Setting: LE_1M, CH12, 0dBm
[13:01:48]  TX_1M[running] DUT: TX, LE_1M, CH12, PER=0/717(0.00%), RSSI(-55)
[13:01:48]  TX_1M[running] TX Setting: LE_1M, CH29, 0dBm
[13:01:49]  TX_1M[running] DUT: TX, LE_1M, CH29, PER=0/861(0.00%), RSSI(-54)
[13:01:49]  TX_1M[   pass] Success
```

Figure 10-16 TX Output Power Status

10.7 RX Sensitivity

RX Sensitivity tests the PER of the DUT. APT transmits a burst of TX on specified `RF channels`, Atmosic Golden IC `AG power level` and `PHY type`, and DUT starts to receive for a certain time. Figure 10-17 shows the RX Sensitivity Setting.

RX sensitivity	Enable
RF channels	(Many items)
AG power level	Level 0 (smallest)
PHY type	1M
Testing time	400 ms
RSSI tolerance lower (dB)	30
RSSI tolerance upper (dB)	30
PER limitation (%)	90
DUT Golden RSSI (dBm)	(Many items)

Figure 10-17 RX sensitivity Setting

If the PER is lower than the value in the setting page, RX sensitivity passes. The default is 90%. Otherwise, it will report failure. Figure 10-18 shows the RX Sensitivity Status.

```
[21:05:41]  RX_1M[running] RX Setting: CH0 Power:Level 1, Path Loss: 1dB
[21:05:42]  RX_1M[running] DUT:RX, LE_1M, CH00, PER=0/1198(0.00%), DUT RSSI(-72).
[21:05:42]  RX_1M[running] RX Setting: CH19 Power:Level 1, Path Loss: 1dB
[21:05:43]  RX_1M[running] DUT:RX, LE_1M, CH19, PER=0/1069(0.00%), DUT RSSI(-73).
[21:05:43]  RX_1M[running] RX Setting: CH39 Power:Level 1, Path Loss: 1dB
[21:05:44]  RX_1M[running] DUT:RX, LE_1M, CH39, PER=7/1213(0.58%), DUT RSSI(-75).
[21:05:44]  RX_1M[   pass] Success
```

Figure 10-18 RX Sensitivity Status

10.8 Function Test

Function test is a test for normal Bluetooth LE operations. There are two types of testing: advertisement and connection by selecting `Advertising` or `Connection` to `True` respectively. Figure 10-19 shows the Function Test Setting.

Function test	Enable
Advertising?	True
Connection?	True
Timeout value	3000

Figure 10-19 Function Test Setting

CONFIDENTIAL

The `Timeout value` is the time within which DUT could complete the operation. Otherwise, it will report failure. This test will launch after the normal firmware is programmed, if the firmware doesn't support advertising this item should not be enabled.

Figure 10-20 shows the Function Test Status.

```
[21:05:44]  Burn_FW_NVDS[running] Burn BLE_adv.elf...
[21:05:44]  Burn_FW_NVDS[running] Burn tmpE9B7.tmp...
[21:05:55]  Burn_FW_NVDS[   pass] Burn success
[21:05:57]  Adv_Test[   pass] ADV test success.
[21:05:58]  Conn_Test[   pass] CONN test success.
```

Figure 10-20 Function Test Status

CONFIDENTIAL

10.9 User Firmware (.atm, .mpbin, .elf) / NVDS file (.bin) / Tags Customization

After selecting the files to the `User firmware (.atm, .mpbin, .elf)` and `NVDS file (.bin)`, AtmMPtool.exe will program flash after all RF related test items have passed. Figure 10-21 shows the User Firmware/Tags Setting.

If you want to use enhancement method in programming FW, please set `True` for `Programming FW speed up.`

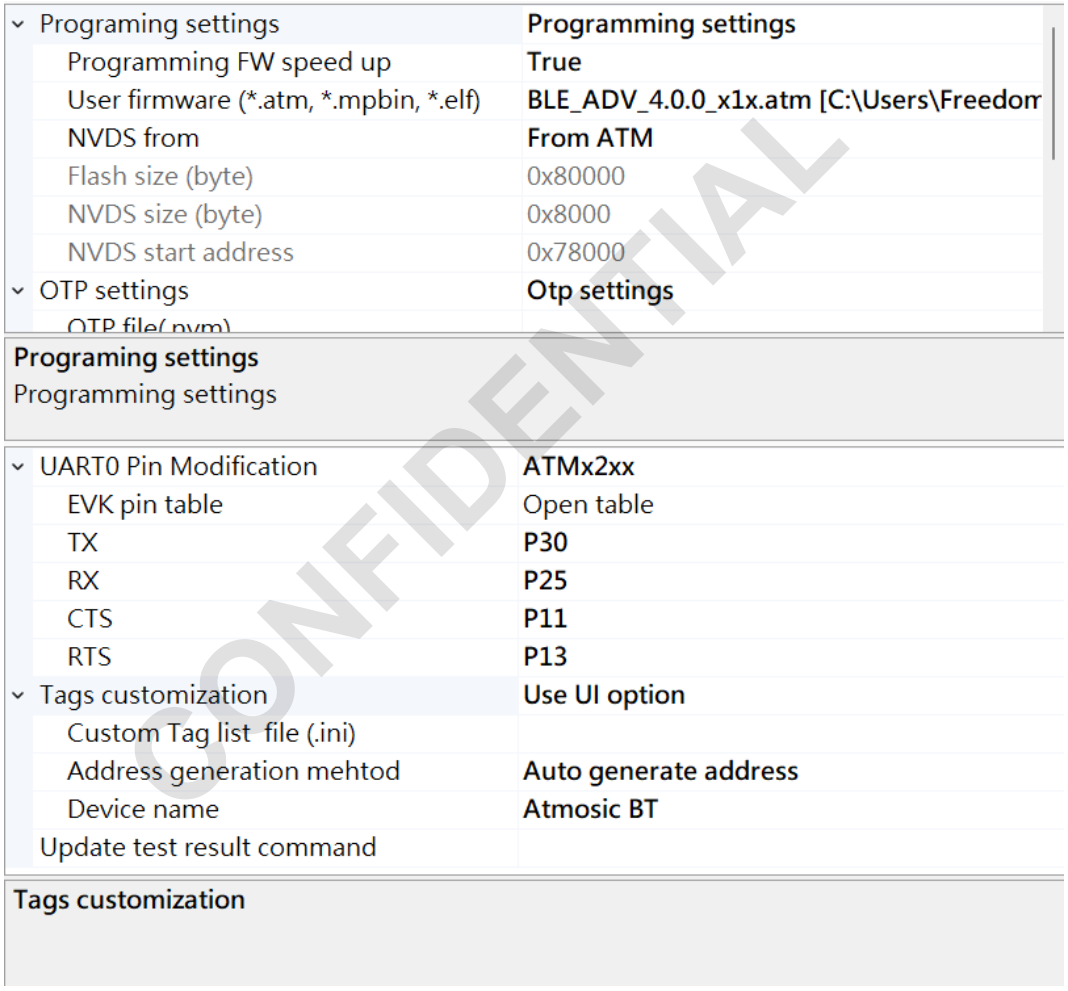


Figure 10-21 User Firmware/Tags Setting

The Tags customization is used to give every device a different Bluetooth address, name (if needed), and application specific tag. For the application specific tag, it could be defined by assigning a Custom Tag List file, using the format:

CONFIDENTIAL

```
<Number of Entry(decimal)> = <Tag(hex)>
<Length(hex)>
0=<Data(hex)>
1=<Data(hex)>
2=...
  ...
```

Figure 10-22 shows 500 entries with tag 0xAA and its data length is 48 bytes (0x30).

```
500 = AA 30
0=e5a72e4629d341389dbe26eef485d08ab643f236356342bfa3b4f63c7cbc6558bf4f7c6a56394a018cfecac61078755f
1=646c921C03ac48bfbab22230fd36760f7e3c32f08226434ea50c3ec8ff6f3b6c32109069ce0042329e70bf4ee8b5a7c8
2=f7791bac37dc4c4ebd8882c160fd17b294e7cf1c52e94642810014ccf33d883ec1873d1ff6214239bfa3ebadc9c1a6f0
3=. . .
```

Figure 10-22 Example of 500 Entries

CONFIDENTIAL

For Bluetooth address tag, there are two kinds of address generation methods: auto generate address and manual assignment.

If auto generate address mode is selected, the address will be generated randomly as a static random address of Bluetooth standard on each device.

If manual assignment is selected, the user needs to define address range as an address pool.

The Bluetooth address pool which is started from `Started Bluetooth address` and has `Total count` number of addresses. These addresses will be assigned to each tested DUT. The `Residue` indicates the number of unused addresses in the pool. The `Current bluetooth address` indicates the next address which will be assigned to the next DUT.

If the pool is empty when the addresses run out, some messages will be shown to the user and AtmMPtool.exe will not assign an address to DUT until the user specifies a new address pool. Figure 10-23 shows an Out of Address Error.

```
[10:55:36] ----->Boot<-----
[10:55:36] Load config file from: C:\Lab\setting.xml
[10:55:36] Factory state = InitMpSoftware
[10:55:36] Factory state = SearchingMpBoard
[10:55:37] Factory state = Wait4Start
[10:55:37] base = AA:BB:CC:DD:EE:FF, total = 1, left = 0
```

Figure 10-23 Out of Address Error

The `Device name` is used to specify the DUT's name. It could be treat with a c# string with an extra parameter of uint64 form of assigned Bluetooth address.

For example, if the current DUT Bluetooth address is 7C:69:6B:00:03:FF and `Device name` is AT-{0:X3}, then the result would be AT-3FF.

10.10 OTP File (.nvm) (ATM2/ATM3 only)

This property is used to assign a specific NVM file to update OTP. After selecting the file, the OTP area will be programmed in the last step of the MPTool process. Usually, the NVM file is used for OTP application or some complement HW parameters setting.

OTP settings	Otp settings
OTP file(.nvm)	
Hardware EFuse (On->Off)	(32K, SWD, 32kCheck) : (On, On, Need)

Figure 10-24 OTP settings

The option "Hardware EFuse" has some options, to modify the hardware config bit in the DUT, this is irreversible operation. If you want to select this option, please check your hardware designer.

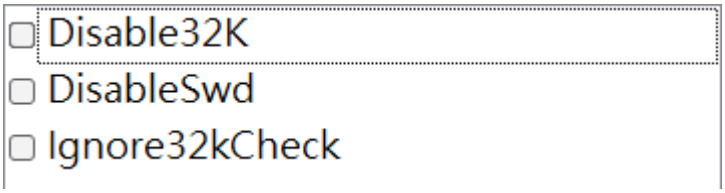


Figure 10-25 Hardware EFuse settings

More calibration data option please see Calibration Settings

CONFIDENTIAL

10.11 UART0 Pin Modification

In the Setting page, under the option UART0 pin Modification, please set the UART pins according to the GPIO used on the DUT. See Figure 10-26.

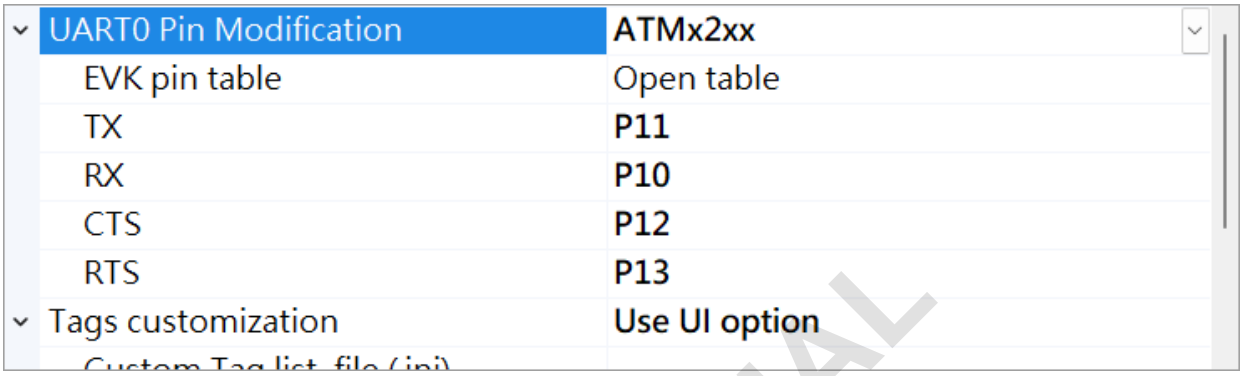


Figure 10-26 UART0 Pin Modification

The option UART0 Pin Modification shows all supported Atmosic devices, you need to choose the device you used, then the pins list will change to the available list.

If your board is Atmosic's EVK or module, you can open EVK pin table, which provides the default mappings, see Figure 10-27.



Figure 10-27 UART Setting for EVK (EVB) and Modules

CONFIDENTIAL

10.12 Start Option

The `Start Option` property is for users to choose how to start the MPTool process. Currently, two options are supported, `Button Start` and `SN Input Start`.

When `Button Start` is selected, the user clicks the `Start/Stop` button to control the testing procedure.

When `SN Input Start` is selected, the user inputs the serial number into the SN text box then press enter to start.

The serial number default is 12 hexadecimal letters. Figure 10-28 shows the selection of start options. Figure 10-29 shows the SN text box which waits for start.

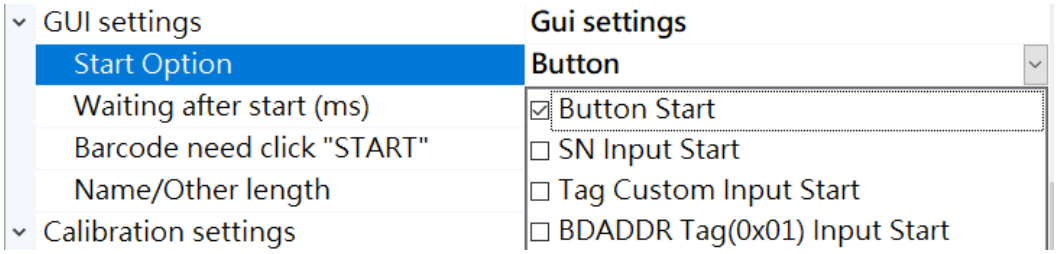


Figure 10-28 Start Option

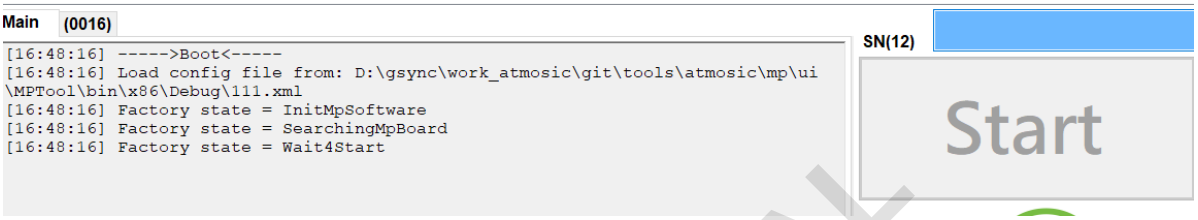


Figure 10-29 SN Text Box

10.13 Write MP Region (This is only supported for ATM2/ATM3)

If your firmware supports calibration data to be written in the MP region (Flash), you can write calibration data in it through the following settings.

When you load an atm file that supports the MP region, you will see the fields in Figure 10-30 display the MP region information corresponding to the atm file.

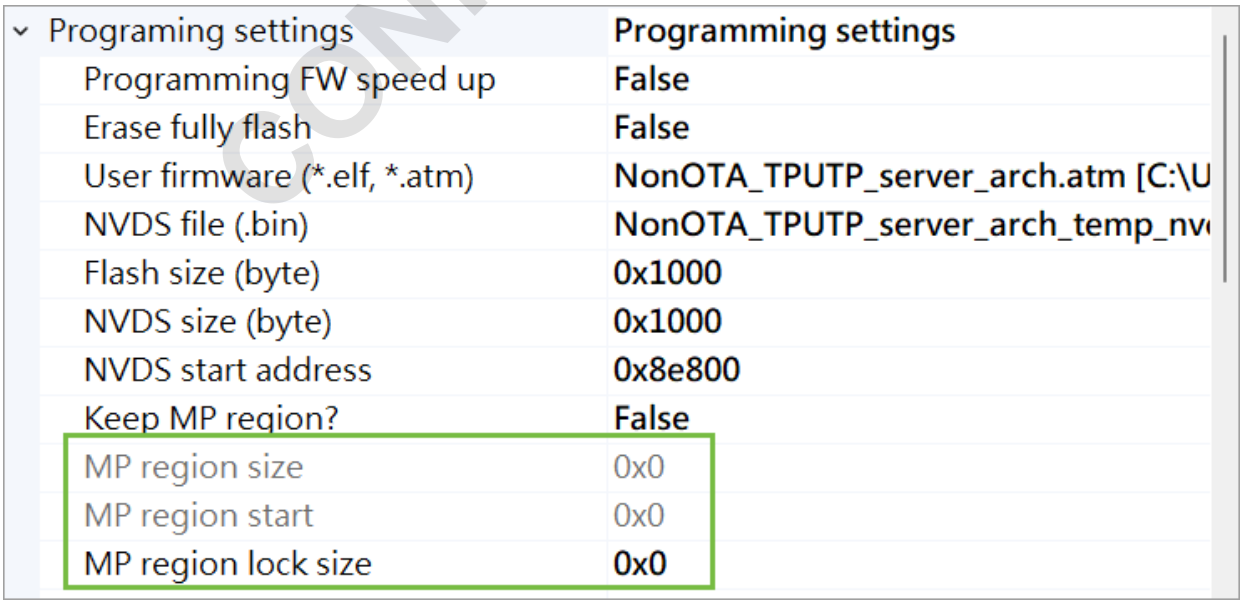


Figure 10-30 MP Region Information

CONFIDENTIAL

You also need to select this option to set up the MPTool process to write calibration data into the MP region. This is only supported with ATM2/ATM3 Wireless SoCs.

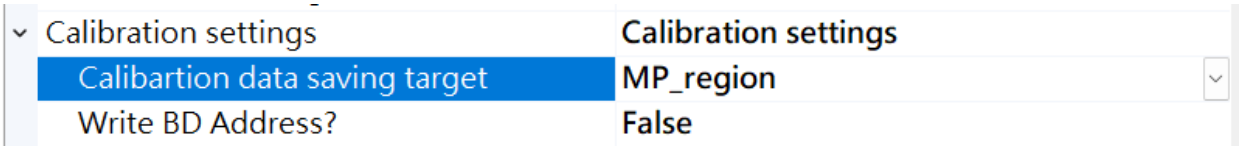


Figure 10-31 Write Calibration Data to MP Region

For more calibration data options, please see Calibration Settings section.

10.14 Command Mode

After clicking the menu item `Command mode`, a console will show up. Figure 10-32 shows the Command Mode Setting.

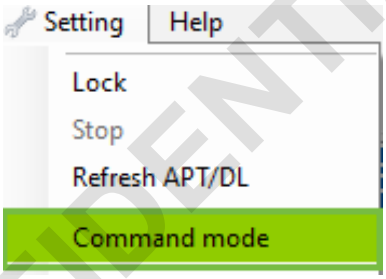


Figure 10-32 Command Mode Setting

By typing any invalid word, the usage will display to guide the user on the available commands. Figure 10-33 shows the Command Menu.

CONFIDENTIAL

```
HELP <MP|DUT|AG|FILE|LOG|SETTING|PARAM>
Usage:

* CLI
    Enter interactive mode.

* APT DISC
    Discover board.

* APT <DevId> (AG|DUT) ...
    Control AG or DUT.

* APT <DevId> MP ...
    Test DUT through AG

* FILE (NVDS|OTP) <file> ...
    Manipulate NVDS or OTP files

* LOG ...
    Control log file setting

* SETTING ...
    Common value Setting

* HELP <MP|DUT|AG|FILE|LOG|SETTING|PARAM>
    Help for detail command descriptions.


================================================================================
MP >
```

CONFIDENTIAL

Figure 10-33 Command Menu

Users can use command mode to test against one pair of APT and DUT.

10.15 PV Harvesting Test

This test is for PV harvesting enabled applications to test its functionality by using harvesting detection signal in the ATM3 or ATM33e. The test set up is illustrated in Figure 10-34.

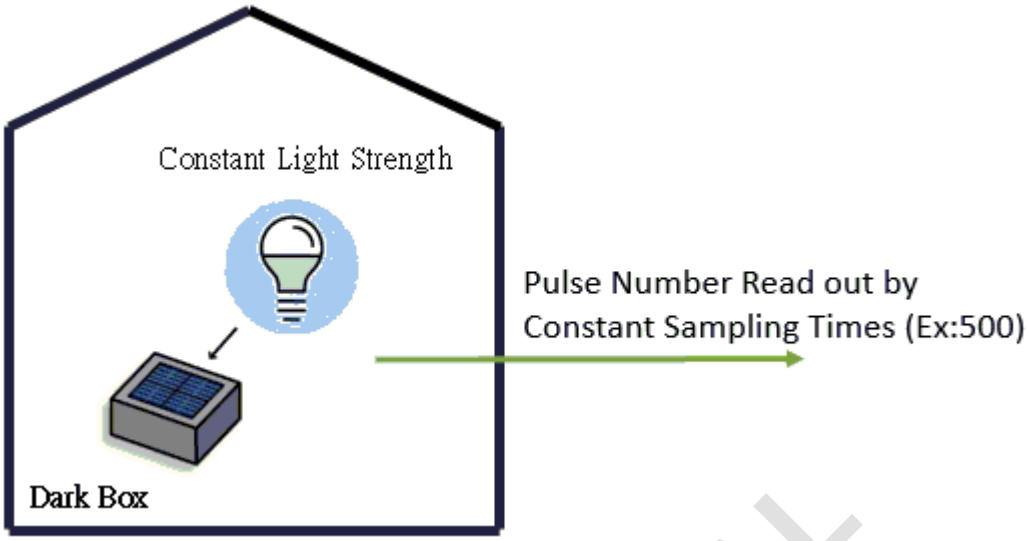


Figure 10-34 PV Test Setup

CONFIDENTIAL

The harvesting detection signal monitors the incoming harvesting power from the VHARV pin and the duty-cycled behavior of the switching supply that supplies power to the chip. This test polls the harvesting detection signal 500 times and records the times it is high to roughly estimate the harvesting power consumed by the chip, which is also an indicator of the harvesting function.

Please note this test only indicates if the chip harvesting function is working. It's not a precise measurement of the actual light level or harvested energy.

Figure 10-35 shows the setting that can be adjusted based on the PV cell model and light level.

PV Test	Enable
Test Times	500
Tolerance Lower	1
Tolerance Upper	500

Figure 10-35 PV Test Setting

- Test Times
  - The number of times that polls the harvesting detection signal.The default value is 500. Please don't change it

- Tolerance Lower
  - Lower limit of the harvesting power
- Tolerance Upper
  - Upper limit of the harvesting power. The maximum value should equal to the value of test times

The tolerance varies from different light levels in production line and also different PV cell models. Atmosic suggests that customers test a batch of the DUT and find the appropriate criteria.

10.16 External NVDS Process

If you want to write some NVDS with custom rules, you can refer to this chapter:

1) Change `Tags customization` to `Call external process`.
2) If you need to get NVDS by a process, set `Get external NVDS command` to the command that gets the NVDS information XML for the MPTool, see Get External NVDS Process.
3) When the update is finished and you want to save which information is used, you can set `Notify external NVDS command` to the command that notifies the NVDS information is used, see Notify External NVDS Used Process.

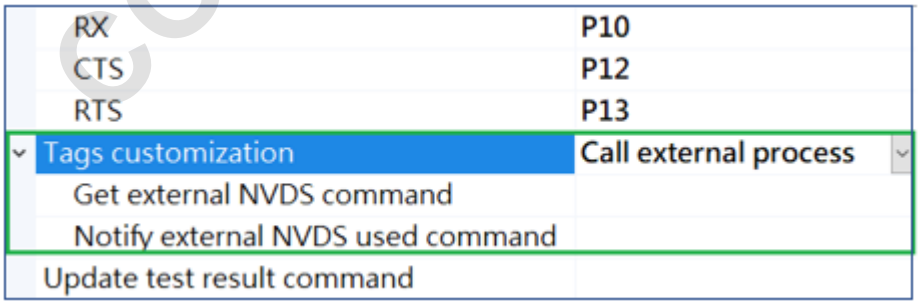


Figure 10-36 Get Custom NVDS with Call External Process

10.16.1 Get External NVDS Process

When you set the command from the MPTool, the MPTool will call it before writing NVDS or OTP with the command:

CONFIDENTIAL

- <the command in option> <the xml path that MPTool want external process write>

If you set the command external.exe arg1 , MPTool will call the process by external.exe arg1 xxxx.xml and expect external.exe will write NVDS information in the xxxx.xml.

Below is the reference of XML format:

```
<root>
  <device index="{index from external process}">
    <NVDS type="Flash" tag="0x{tag1 hex}">{data of tag1}</NVDS>
    <NVDS type="OTP" tag="0x{tag2 hex}">{data of tag2}</NVDS>
  </device>
</root>
```

CONFIDENTIAL

10.16.2 Notify External NVDS Used Process

When you set the command for the MPTool, the MPTool will call it after writing NVDS and OTP with the command:

- <the command in option> <the index in the XML created when called get external NVDS command> <PASS or FAIL>

If you set the command external.exe arg2 , MPTool will call the process by external.exe arg2 <index> `PASS` when write information success and external.exe arg2 <index> `FAIL` when write information failure.

10.17 Update Test Result with External Process

After the testing is finished, MPTool will write results into the .csv file (refer to Atmosic Mass Production Tool User Interface) for each DUT tested, and you also can use this feature to update this information to your external process.

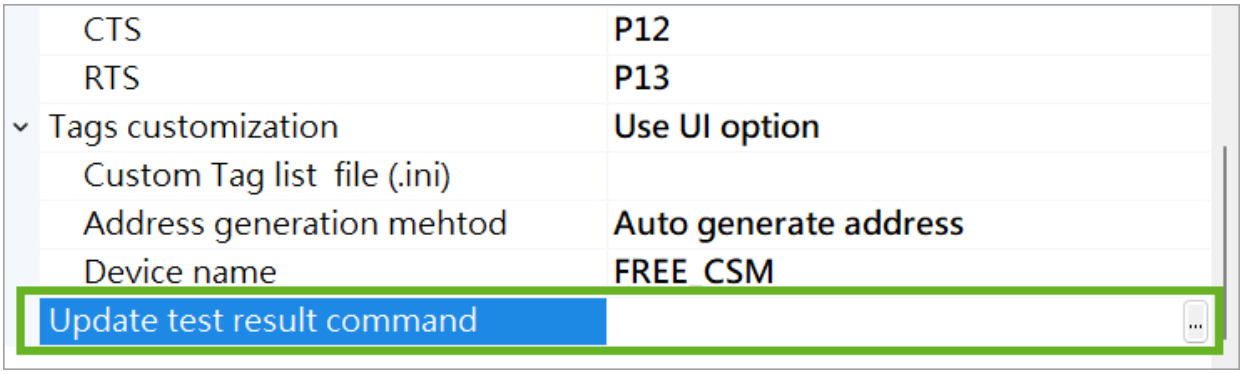


Figure 10-37 Update Test Result with Call External Process

If you set a command in this option, when testing finished, MPTool will call the process with the command:

- <the command in option> <.csv file>

The .csv file is the test report created from DUT testing. This feature allows users to develop a process to analyze the test results after each test, and store it in the users' database.

10.18 Calibration Settings

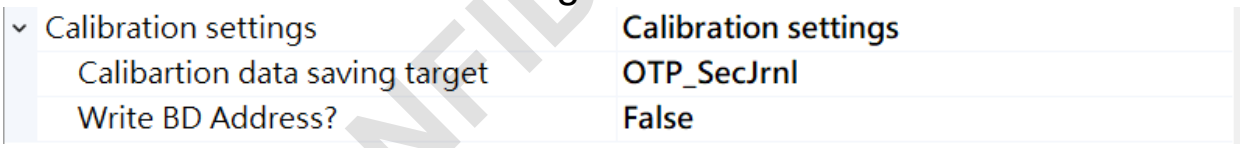


Figure 10-38 Calibration settings

The `Calibration data saving target` for choosing saving calibration data to OTP, Secure Journal or MP Region.

Set `Write BD Address` to true will write BD address to the calibration data areas.

If you want to write Crystal CAP, please see Crystal Trim.

11. Atmosic Production Tool Console Mode

The Atmosic Mass Production Tool also provides console mode for user control. The user needs to go to the working directory in C:\AtmosicMP\bin and input the command. User should follow the command format `AtmMPtool.exe APT` parameters.

There are a few conventions in these command formats:

- S symbol is a fixed string S. The matched input could be case-insensitive.

CONFIDENTIAL

- <S> symbol is for dynamic inputs, such as an ID or a file name. The matched input would depend on the situation of the real execution environment.
- (S1|S2|...|Sn) symbol is one of the fixed strings between S1 to Sn. The matched input shall be one of S1 to Sn being case-insensitive.
- [S] symbol is for enumerated number. Only predefined numbers are allowed.
- <<S>> symbol is optional and the definition is the same as <S>.
- [[S]] symbol is optional and the definition is the same as [S].

Before OTP and NVDS programming, users may need to prepare an updated binary file. The Atmosic Mass Production Tool provides a tool named nvds_tool.exe for users to manipulate the tags and form the binary file. Please refer to the OTP/NVDS/MP Region File Content Display section for the usage.

CONFIDENTIAL

11.1 Windows System Environment Variable

This chapter shows the Windows operating system environment variables used to adjust the CLI test.

Variable name	Description
SKIP_DETAIL_LOG	If you want to simplify the log generated by the CLI in the console (cmd.exe), you can set this environment variable to 1. (e.g., set SKIP_DETAIL_LOG=1)

11.2 Board Discovery

Discover the connected APT/DL boards by using the following command.

Command
• AtmMPTool APT DISC

Parameter	
DISC	Discover APT/DL boards
Result	Return all <DevID> of APT/DL boards

11.3 DUT/AG Tx Test

Apply the APT (also known as AG) or DUT to Bluetooth LE Tx mode.

Command	
• AtmMPTool APT <DevID> (AG|DUT) TEST TX [PHY] [CHNL] [PKTYPE] <PktLen> <Tms>	
Parameter	
(AG|DUT)	AG - Atmosic Golden (Not allowed with DL board) DUT - Device under test
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15 4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 255
<Tms>	Time in ms for continuously transmitting packets 0: Infinite 1 ~ 8000
Result	Return how many packets transmitted

***Note**: Users must download HCI firmware to RAM prior to using this command with DUT (AG|DUT).*

CONFIDENTIAL

11.4 DUT/AG Rx Test

Apply the APT (also known as AG) or DUT to Bluetooth LE Rx mode.

Command	
• AtmMPTool APT <DevID> (AG|DUT) TEST RX [PHY] [CHNL] [PKTYPE] <PktLen> <Tms> • AtmMPTool APT <DevID> (AG|DUT) TEST END	
Parameter	
(AG|DUT)	AG - Atmosic Golden (Not allowed with DL board) DUT - Device under test
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)

[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15 4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 255
<Tms>	Any number from 1 to 8000 but cannot set 0
Result	Return how many packets received

***Note**: Users must download HCI firmware to RAM prior to using this command with DUT (AG|DUT).*

CONFIDENTIAL

11.5 DUT/AG Tx Power Gain

Command and parameters for setting the Tx power to APT (also known as AG) or DUT.

Command
• AtmMPTool APT <DevID> (AG|DUT) GAIN [GAINIDX]

Parameter	
(AG|DUT)	AG - Atmosic Golden (Not allowed with DL board) DUT - Device under test
[GAINIDX]	0 - (-20dBm) 1 - (-10dBm) 2 - (-6dBm) 3 - (-4dBm) 4 - (-2dBm) 5 - 0dBm 6 - 2dBm 7 - 4dBm
Result	SUCCESS or FAIL

***Note**: Users must download HCI (DTM) firmware to RAM prior to using this command with DUT role.*

11.6 DUT RAM or OTP Programming

Program DUT RAM FW or DUT OTP data.

Notice: If you downloaded default RAM code and want to use UART, please call UART rx enable command or change RX pin mux first.

Command
• AtmMPTool APT <DevID> DUT FLASH (RAM|OTP|POTP|NSWD|N32C) <file>

Parameter	
(RAM|OTP|POTP|N32C)	RAM - Program image to RAM for RF testing OTP - Program OTP data to OTP **(CANNOT REVERT SETTING!)** POTP - Pull OTP data from OTP NSWD - SWD connection will be prohibited after setting **(CANNOT REVERT SETTING!)** N32C - No 32 kHz crystal on board **(CANNOT REVERT SETTING!)**
<file> Optional for RAM Not used in N32C	For RAM: If not provided, C:\AtmosicMP\fw\DTM.bin will be used For OTP: Read this file and program to OTP For POTP: Pull data from OTP and write to this file
Result	SUCCESS or FAIL

CONFIDENTIAL

11.7 DUT Firmware Programming

Program firmware to flash.

Command
• AtmMPTool APT <DevID> DUT FLASH FW <atm file> [EUD] [NCE] • AtmMPTool APT <DevID> DUT FLASH FFW <atm file> [EUD] [NCE] • AtmMPTool APT <DevID> DUT FLASH FW <elf file> <addr> <size> [EUD] [NCE]

Parameter	
FW, FFW	FW - Program firmware and verification. FFW - Program firmware and verification. And the program process is using speed up process.
<atm file>, <elf file>	File pathname (.elf, .atm) For Example, C:\AtmosicMP\Test.elf for firmware
<addr> Optional	FW start address This information is customer specific If not provided, the default address of FW is 0x0
<size> Optional	FW maximal size This information is customer specific

	If not provided, the default maximum size of FW is 0x80000 (512 KB)
[EUD] Optional	Erase upgrade data or not 1 for erasing If not provided, the default value is 1
[NCE] Optional	Do chip erase before programing flash
Result	SUCCESS or FAIL

11.8 DUT NVDS Programming or Pull

Program NVDS data to flash (for ATM2/ATM3).

Command
• AtmMPTool APT <DevID> DUT FLASH (NVDS|PNVDS) <file> <addr> <size> • AtmMPTool APT <DevID> DUT FLASH (NVDS|PNVDS) <file>

Parameter	
(NVDS|PNVDS)	NVDS - Program NVDS data to flash PNVDS - Pull nvds data from FLASH
<file>	NVDS file pathname (.bin) For Example,C:\AtmosicMP\flash_nvds.bin
<addr> Optional	NVDS start address This information is customer specific If not provided, the default address of NVDS is 0x78000
<size> Optional	NVDS maximal size This information is customer specific If not provided, the default maximum size of NVDS is 0x8000 (32 KB)
Result	SUCCESS or FAIL

CONFIDENTIAL

11.9 DUT Firmware Validation

Validate firmware in flash.

Command
• AtmMPTool APT <DevID> DUT FLASH VFW <file> <addr> • AtmMPTool APT <DevID> DUT FLASH VFW <file>

Parameter	
<file>	File to be programmed. ELF file for firmware For Example, C:\AtmosicMP\Test.elf
<addr> optional	FW start address This information is customer specific If not defined, the default address of FW is 0x0
Result	SUCCESS or FAIL

11.10 DUT Firmware and NVDS Programming Concurrently

Program firmware and NVDS data to flash at the same time.

Command
• AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN> <FADR> <FSIZE> <NADR> <NSIZE> [EUD] • AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN> <FADR> <FSIZE> <NADR> <NSIZE> • AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN> <FADR> <FSIZE> • AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN>

Parameter	
<ELF>	FW file (.elf) to be programmed For Example, C:\AtmosicMP\Test.elf
<BIN>	NVDS file (.bin) to be programmed For Example, C:\AtmosicMP\flash_nvds.bin
<FADR> Optional	FW start address This information is customer specific If it is not defined, the default value is 0x0
<FSIZE> Mandatory, if set FADR	FW max size This information is customer specific If it is not defined, the default value is 0x80000 (512 KB)
<NADR> Optional	NVDS start address This information is customer specific

CONFIDENTIAL

	If it is not defined, the default value is 0x78000
<NSIZE> Mandatory, if set ADDR	NVDS max size This information is customer specific If it is not defined, the default value is 0x8000 (32 KB)
[EUD] Optional	Erase upgrade data or not 1 for erasing If not provided, the default value is 1
Result	SUCCESS or FAIL

11.11 DUT Flash Erase

Program firmware and NVDS data to flash at same time.

Command
• AtmMPTool APT <DevID> DUT FLASH ERASE <FADR> <FSIZE>

Parameter	
<FADR>	Start address If not provided, default is 0
<FSIZE>	Erase Size If not provided, the default value will be the flash size
Result	SUCCESS or FAIL

CONFIDENTIAL

11.12 DUT/AG Reset Command

Reset APT (also known as AG) or DUT.

Command
• AtmMPTool APT <DevID> (AG|DUT) RESET

Parameter	
(AG|DUT)	AG - Atmosic Golden (Not allowed with DL board) DUT - Device under test
Result	SUCCESS or FAIL

11.13 DUT/AG Version

Command to query the chip version of APT (also known as AG) or DUT.

Command
• AtmMPTool APT <DevID> (AG|DUT) VERSION

Parameter	
(AG|DUT)	AG - Atmosic Golden (Not allowed with DL board) DUT - Device under test
Result	Show Atmosic device version of AG or DUT

***Note**: Users must download HCI firmware to RAM first prior to using this command with the DUT role.*

11.14 DUT/AG NVDS Read

Read NVDS tag data.

Command
• AtmMPTool APT <DevID> (AG|DUT) NVDS READ <TAG>

Parameter	
(AG|DUT)	AG - Atmosic Golden (Not allowed with DL board) DUT - Device under test
<TAG>	Decimal value from 0 to 255 Please get this value from RD
Result	Show the NVDS data

11.15 DUT Log

Command to show the DUT log after DUT power on.

Command
• AtmMPTool APT <DevID> DUT LOG

Parameter	

CONFIDENTIAL

No parameters	
Result	Show DUT log

11.16 Bluetooth LE RF PER Test (Packet Error Rate)

Command for packet error rate testing.

Command
• AtmMPTool APT <DevID> MP TEST PER [PHY] [CHNL] [PKTYPE] <PktLen> <Tms>

Parameter	
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15 4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 63
<Tms>	Set the PER test duration 200 ~ 8000
Result	Show the PER test result with both directions

***Note**: Users must download HCI firmware to RAM prior to using this command.This command is only allowed with the APT board, not with DL board.*

CONFIDENTIAL

11.17 Bluetooth LE RF FOS Test (Frequency Offset)

Command for frequency offset testing.

Command
• AtmMPTool APT <DevID> MP TEST FOS [PHY] [CHNL] [PKTYPE] <PktLen>

Parameter	
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15 4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 63
Result	Show the FOS test result with both directions

***Note**: Users must download HCI firmware to RAM prior to using this command. This command is only allowed with the APT board, not with DL board.*

CONFIDENTIAL

11.18 Bluetooth LE RF CAL Test (Calibration)

Use APT to calibrate M crystal.

Command
• AtmMPTool APT <DevID> MP TEST CAL [GAINIDX] [CHNL]

Parameter	
[GAINIDX]	Tx power gain: 0 - (-20 dBm) 1 - (-10 dBm) 2 - (-6 dBm) 3 - (-4 dBm) 4 - (-2 dBm)

	5 - 0 dBm 6 - 2 dBm 7 - 4 dBm
[CHNL]	BLE Channel: 0 ~ 39
Result	Show the calibration result

***Note**: Users must download HCI firmware to RAM prior to using this command. This command is only allowed with the APT board, not with DL board.*

11.19 OTP/NVDS/MP Region File Content Display

This command shows the content of OTP (.nvm) and NVDS (.bin) files.

Command
• AtmMPTool FILE (OTP | NVDS | MPREGION) <file> SHOW

Parameter	
(OTP | NVDS | MPREGION)	OTP - OTP file type NDVS - NVDS file type MPREGION - MP region file type
<file>	File path which used to be read and displayed
Result	Shows the content of the file with rows of tags

11.20 NVDS/MP Region Create Blank File

This command can create NVDS (.bin) and MP Region (.bin) files.

Command
• AtmMPTool FILE (NVDS | MPREGION) <file> BLANK

Parameter	
(NVDS | MPREGION)	NDVS - NVDS file type MPREGION - MP region file type
<file>	File path used to be read and displayed
Result	Shows the content of the file with rows of tags

CONFIDENTIAL

11.21 OTP/NVDS/MP Region Tag Addition with File

This command adds specific TAG and its data from content of OTP (.nvm) and NVDS (.bin) file and output to a new file.

For Tag ID please refer to section Common OTP/NVDS/MP Region Tag ID.

Command
• AtmMPTool FILE (OTP | NVDS | MPREGION) <file> TAG ADD <tag> <hex data> <new file>

Parameter	
(OTP | NVDS | MPREGION)	OTP - OTP file type NDVS - NVDS file type MPREGION - MP region file type
<file>	File path of original file
<tag>	Tag number Hex format, e.g, 01, 45, ...
<hex data>	Data with hex format, e.g, 6 bytes data would be input by 112233445566
<new file>	File path to save
Result	<new file> will be saved and shows success

11.22 OTP/NVDS/MP Region File Merge

This command merges two OTP (.nvm) or NVDS (.bin) files.

Command
• AtmMPTool FILE (OTP | NVDS | MPREGION) <file> MERGE <add file> <new file>

Parameter	
(OTP|NVDS|MPREGION)	OTP - OTP file type NDVS - NVDS file type MPREGION - MP region file type
<file>	File path of original file
<add file>	File path of added file
<new file>	File path to save
Result	<new file> will be saved and shows success

CONFIDENTIAL

11.23 DUT/AG Test ADV

Start or stop advertising

Command
• AtmMPTool APT <DevID> (AG|DUT) TEST ADV START <payloads> • AtmMPTool APT <DevID> (AG|DUT) TEST ADV STOP

Parameter	
(START|STOP)	Testing start or stop
<payloads>	Use --0x<tag>.(ascii|hex)=data to set payload context E.g. --0x09.ascii=001122334455
Result	Show the ADV command working status

11.24 DUT/AG Test WuRx

WuRx test commands.

Command
• AtmMPTool APT <DevID> (AG|DUT) TEST WURX SEND_TX_PACKET • AtmMPTool APT <DevID> (AG|DUT) TEST WURX RESET_RX • AtmMPTool APT <DevID> (AG|DUT) TEST WURX CHECK_RX_COUNT <count1_limit> <count2_limit>

Parameter	
(SEND_TX_PACKET | RESET_RX | CHECK_RX_COUNT)	SEND_TX_PACKET: send WURX waveform RESET_RX: reset device to catch WuRx packet CHECK_RX_COUNT: check WuRx packet count
<count1_limit>, <count2_limit>	set limit for count
Result	Show the WuRx command working status and register change log

CONFIDENTIAL

11.25 DUT/AG I/O Setting

Control GPIO pin action or get GPIO pin status.

Command
• AtmMPTool APT <DevID> (AG|DUT) IO (SETUP|SET) <io> <0|1> • AtmMPTool APT <DevID> (AG|DUT) IO (GET) <io>

Parameter	
(SETUP|SET|GET)	SETUP: set I/O pin to input(0) or output(1) SET: set output I/O pin to low(0) or high(1) GET: get I/O pin low or high
<io>	number of pin
Result	Show the I/O setting command working status

11.26 DUT/AG Set/Get Runtime Crystal CAP

Set or get runtime crystal CAP value.

Command
• AtmMPTool APT <DevID> (AG|DUT) CAP SET <val> • AtmMPTool APT <DevID> (AG|DUT) CAP GET

Parameter	
(SET | GET)	SET: set output I/O pin to low(0) or high(1) GET: get I/O pin low or high
<val>	0 ~ 31
Result	Show the CAP value setting command working status

CONFIDENTIAL

11.27 Set UART Rx Enable

When you download default RAM code, and need to use UART, please call this command.

Command
• AtmMPTool APT <DevID> DUT UART RX_ENABLE

Parameter	
No parameters	
Result	Show the UART Rx enable command working status

11.28 Change the UART Pin Mux

If your product's UART pin definition does not equal the default setting, please use this command to change it before calling UART RX enable.

Command
• AtmMPTool APT <DevID> DUT UART MUX <Tx pin> <Rx pin> <Cts pin> <Rts pin>

Parameter	
<Tx pin><Rx pin> <Cts pin><Rtx pin>	Please key in I/O number
Result	Change the UART pin mux setting command working status

11.29 Set 16 MHz Crystal Capacitor

When you need to set capacitor value into your OTP or NVDS file.

For ATM2/ATM3:

You can use this powerful command to replace OTP/NVDS/MP region Tag addition with File.

CONFIDENTIAL

Command
• AtmMPTool FILE (OTP|NVDS) <file> TAG SET_CAP <value> <new file>

Parameter	
<file>	File path of original file
<value>	0~31
<new file>	File path to save
Result	<new file> will be saved and shows success

For ATM33/e:

You can use this command.

Command
• AtmMPTool FILE SECJRNL <file> TAG SET_CAP <value> <new file>

Parameter	
<file>	File path of original file
<value>	0~255
<new file>	File path to save
Result	<new file> will be saved and shows success

CONFIDENTIAL

11.30 Set Testing Log Refresh

The production test will write detailed logs in the log folder. Use this command to create a new log file of the current time and import subsequent logs into the file.

Command
• AtmMPTool LOG REFRESH

Parameter	

No parameters	
Result	the next log will be written to the new log file

11.31 Set Testing Log Finished

Call this command to update the final status on the file name of the log.

Command
• AtmMPTool LOG FINISHED

Parameter	
No parameters	
Result	update log file name if this testing has an error message

11.32 CLI Detail Log

The MPTool CLI mode displays simple test results in the standard output. If you need a more complete test LOG, you can find the (DevID)_CLI_<test_time>.xml file in logs/xml/ under the same directory of AtmMPTool.exe, which contains a more detailed log.

You can use Set Testing Log Refresh and Set Testing Log Finished to cut the log into different files according to the test flow.

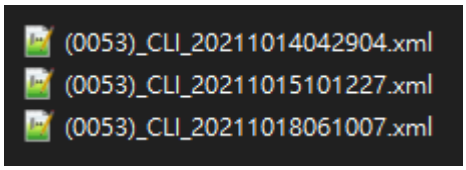


Figure 11-1 CLI Command Detail Logs

11.33 Write MP Region

Call this command to update the final status on the file name of the log.

Command
• AtmMPTool APT <DevID> DUT FLASH MPREGION <mp_region_file> <mp_region_addr> <mp_region_max_size> <mp_region_lock_size>

Parameter	

CONFIDENTIAL

<mp_region_file>	bin file to be programmed
<mp_region_addr>	start address for mp region
<mp_region_max_size>	max size for mp region
<mp_region_lock_size>	protection size for mp region
Result	The binary content of the MP region file will be written to the specified address

***Note**: The BD address tag in MP region would take effect during firmware initialization period.*

11.34 Pull MP Region

Call this command to pull the binary of the MP region from flash in the IC.

Command
• AtmMPTool APT <DevID> DUT FLASH PMPREGION <mp_region_file> <mp_region_addr> <mp_region_max_size>

Parameter	
<mp_region_file>	bin file to be programmed
<mp_region_addr>	start address for mp region
<mp_region_max_size>	max size for mp region
Result	Save the binary content of the MP region in the IC's flash to the file

11.35 Set UART Baud Rate

Call this command to set the baud rate to the CLI common setting, this setting will continue until the next time this command is called again.

Command
• AtmMPTool SETTING BAUDRATE <baud rate>

Parameter	
<baud rate>	UART baud rate
Result	Update global setting

CONFIDENTIAL

11.36 Dump Flash to Bin File

Dump flash to a bin file.

Command
• AtmMPTool APT <DevID> DUT FLASH DUMP <bin file> (<addr>) (<size>)

Parameter	
<bin file>	output file name
<addr>	start address (default is 0x0)
<size>	size (default is dump to end)
Result	Save the binary content of the IC flash to a file

11.37 Flash Chip Erase

Erase all contents of flash.

Command
• AtmMPTool APT <DevID> DUT FLASH CHIP_ERASE

Parameter	
No parameters	
Result	Erase all contents of flash

11.38 Check Flash Protection (Only for ATM2/ATM3)

Check if the flash is protected or not.

Command
• APT <DevID> DUT FLASH CHECK_PROTECTING

Parameter	
No parameters	
Result	If it fails, show error status and its error reason

CONFIDENTIAL

11.39 Show ATM File Information

Show the information from ATM file.

Command
• AtmMPTool FILE ATM <file> SHOW

Parameter	
<file>	Input atm file path
Result	Display the information on the console

11.40 Export FW Bin File from ATM File

Export FW bin file from ATM file.

Command
• AtmMPTool FILE ATM <in_file> EXP_FW <out_file>

Parameter	
<in_file>	input atm file path
<out_file>	output bin file path
Result	Export FW bin to file path

11.41 Export NVDS Bin File from ATM File

Export NVDS bin file from ATM file.

Command
• AtmMPTool FILE ATM <in_file> EXP_NVDS <out_file>

Parameter	
<in_file>	input atm file path
<out_file>	output bin file path
Result	Export NVDS bin to file path

CONFIDENTIAL

11.42 FTDI Pin Control

Command
• AtmMPTool APT <DevID> AG FTDI_CONTROL (NO_POWER) (<CONTROL_FLOW...>)

Parameter	
NO_POWER	If set this parameter, default VBAT, VCCIO, LS pin would set to low
<CONTROL_FLOW…>	Follow the format <ACTION_LABLE> <delay_ms> <ACTION_LABLE> <delay_ms> <ACTION_LABLE>....to control these pins The valid ACTION_LABLE: • VBAT_H : control VBAT pin to high • VBAT_L : control VBAT pin to low • VCCIO_H : control VCCIO pin to high • VCCIO_L : control VCCIO pin to low • LS_H : control LS pin to high • LS_L : control LS pin to low • BBOOT_H : control BBOOT pin to high • BBOOT_L : control BBOOT pin to low • PWD_H : control PWD pin to high • PWD_L : control PWD pin to low Example 1 - sends AtmMPTool APT 50 AG FTDI_CONTROL NO_POWER VBAT_H 50 VCCIO_H 150 LS_H to: 1. Default VBAT, VCCIO, LS are low 2. Set VBAT to high 3. delay 50 ms 4. Set VCCIO to high 5. delay 150 ms 6. Set LS to high Example 2 sends AtmMPTool APT 50 AG FTDI_CONTROL BBOOT_H 20 PWD_H 0 PWD_L 20 BBOOT_L to: 1. Default VBAT, VCCIO, LS are high 2. Set BBOOT to high 3. delay 20 ms 4. Set PWD to high 5. delay 0 ms 6. Set PWD to low 7. delay 20 ms 8. Set BBOOT to low
Result	SUCCESS or FAIL

CONFIDENTIAL

11.43 DUT RRAM Firmware Programming (Only For ATM33/e)

Program firmware to RRAM.

Command	
• AtmMPTool APT <DevID> DUT RRAM FW <atm file> • AtmMPTool APT <DevID> DUT RRAM FW <mpbin file>	
Parameter	
FW	FW - Program firmware and verification
<atm file> / <mpbin file>	File pathname, there has two format supported: 1. *.atm Created from SDK command `make build_archive` 2. *.mpbin The MP bin file context is: —---------- [files] app=<elf or bin file path of app> app_start=<start address of app, if use elf please set 0> bootloader=<elf or bin file path of bootloader> bootloader_start=<start address of bootloader, if use elf please set 0> —---------- You can refer the file C:\AtmosicMP\fw\MP_33xx-5.mpbin
Result	SUCCESS or FAIL

CONFIDENTIAL

11.44 DUT RRAM NVDS Programming (Only For ATM33/e)

Program NVDS data to RRAM.

Command	
• AtmMPTool APT <DevID> DUT RRAM NVDS <file> <addr> • AtmMPTool APT <DevID> DUT RRAM NVDS <file>	
Parameter	
<file>	NVDS file pathname (.bin). E.g, C:\\AtmosicMP\\flash_nvds.bin
<addr> Optional	NVDS start address This information is customer specific If not provided, the default address of NVDS is 0x78000
Result	SUCCESS or FAIL

11.45 DUT Secure Journal Dump or Push (Only For ATM33/e)

Program NVDS data to flash.

Command
• AtmMPTool APT <DevID> DUT SECJRNL DUMP <file> • AtmMPTool APT <DevID> DUT SECJRNL PUSH <file>

Parameter	
<file>	NVDS file pathname (.bin). E.g, C:\\AtmosicMP\\flash_nvds.bin
Result	SUCCESS or FAIL

11.46 DUT RRAM Erase (Only For ATM33/e)

Erase data to 0xff by address, size or all size for the RRAM.

Command
• AtmMPTool APT <DevID> DUT RRAM ERASE <ADDR> <SIZE>

Parameter	
<ADDR>	Start address
<SIZE>	Erase Size If not provided, the default value will be the flash size
Result	SUCCESS or FAIL

CONFIDENTIAL

11.47 DUT Get PV Count

Poll the harvesting detection signal according to the number of test times and return the times it is high to roughly estimate the harvesting power consumed by the chip.

Command
• AtmMPTool APT <DevID> DUT PV_GET <TEST TIMES>

Parameter	
<TEST TIMES>	Test times(500~65535)
Result	PV count value(0~65535)

12. Multiple DUT Testing in CLI Mode

The Atmosic CLI command can execute testing or programming a maximum of eight devices at the same time.

Multiple DUT testing under CLI mode:

```
C:\AtmosicMP\bin>AtmMPTool.exe APT 89 DUT FLASH RAM
RUN "get_rom_version" script
========================================================================
Success
========================================================================
```

```
C:\AtmosicMP\bin>AtmMPTool.exe APT 40 DUT FLASH RAM
RUN "get_rom_version" script
========================================================================
Success
========================================================================
```

CONFIDENTIAL

13. Power Control

13.1 Using the MPTool GUI

The GUI power control is automatic. The GUI options and configuration are similar to previous versions.

13.2 Using the MPTool Console Mode

13.2.1 Using Download Board

The download board DL_V5 (or later), and the MPTool version v2.0.14 (or later) supports DUT's power control. It is important to update the following 3 processes:

- On application opening

  Run this command before setting the DUT on the fixture

  Set all FTDI pins to low (same as power-off)

```
CMD: AtmMpTool.exe APT <DEVID> AG FTDI_CONTROL LS_L 450 VCCIO_L 50 VBAT_L
```

- Before testing

  Change FTDI pins to trigger DUT power-up

```
CMD: AtmMpTool.exe APT <DEVID> AG FTDI_CONTROL VBAT_H 50 VCCIO_H 450 LS_H
```

- After testing

  Set all FTDI pins to low

```
CMD: AtmMpTool.exe APT <DEVID> AG FTDI_CONTROL LS_L 450 VCCIO_L 50 VBAT_L
```

For more information about the FTDI pins control command, refer to FTDI pin control section.

13.2.2 Using APT Board

When using the APT board, the power control is controlled by the pins of the AG.

- On application opening

  Run this command before setting the DUT on the fixture

  Set all FTDI pins to low (same as power-off)

CONFIDENTIAL

```
CMD:
AtmMPTool.exe APT %DevID% AG RESET
AtmMPTool.exe APT %DevID% AG IO SETUP 17 1
AtmMPTool.exe APT %DevID% AG IO SETUP 18 1
AtmMPTool.exe APT %DevID% AG IO SET 17 0
AtmMPTool.exe APT %DevID% AG IO SET 18 0
```

- Before testing

  Change FTDI pins to trigger DUT power-up

```
CMD:
AtmMPTool.exe APT %DevID% AG IO SET 17 1
AtmMPTool.exe APT %DevID% AG IO SET 18 1
```

- After testing

  Set all FTDI pins to low

```
CMD:
AtmMPTool.exe APT %DevID% AG IO SET 17 0
AtmMPTool.exe APT %DevID% AG IO SET 18 0
```

CONFIDENTIAL

14. NVDS Tool

Apart from the FILE command in AtmMptool.exe, NVDS tool provides another way to modify NVDS tags and generate the files which are used for Flash NVDS / OTP or MP region. The executable file is located in C:\AtmosicMP\bin\nvds_tool.exe .

Users may use this tool for three types of operations. See Table 14-1.

Display tags of NVDS or OTP file	
-r <file>	Add from file. For OTP, the -i should be added in front of -r .
-v	Verbose output (with values of tags and data)
-p	Pretty output (with readable meaning of tags and data)
-b	Binary output
-i	Inverted. Used for OTP file, e.g., nvds_tool -i -t otp_nvds.nvm
Add, erase, or lock tags	
-t <tag>	Add tag, e.g., nvds_tool -t 01 -d addr.bin for adding BDADDR tag.
-d <file>	Add data associated with the tag. This option should follow the -t tag.
-e <tag>	Erase tag.
-l <tag>	Lock tag. Once a tag is locked in file , it cannot be unlocked through this tool.
Generation of new NVDS or OTP file	
-o <file>	Output to file. Only allowed with -b output options.

Table 14-1 nvds_tool Options List

CONFIDENTIAL

14.1 DISPLAY Content of NVDS and OTP file

There are four NVDS/OTP file output display formats.

If no output option applied, the default output would be ASCII hex digitals serially.

NVDS display with ASCII output for Flash/RRAM NVDS:

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin
4e
56
44
53
02
06
0a
41
74
6d
6f
73
```

NVDS display with ASCII output for OTP NVDS:

```
C:\AtmosicMP\bin>nvds_tool.exe -i -r otp_nvds.bin
b1
a9
bb
ac
4e
fb
ef
f1
24
f8
d0
2d
```

CONFIDENTIAL

With the binary -b option, the output would be binary data serially.

Binary output is also used to save a new file. Please refer to the section Generate new OTP/NVDS/MP region file for more information.

***Note**: The garbled characters shown above can be ignored as they are caused by the NVDS Tool forcing the use of ASCII code to display the byte array.*

With the verbose -v option, the output would be the ASCII HEX value of each tag and data with address offset.

NVDS display with Verbose output for Flash/RRAM NVDS:

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin -v
@0007 02  41 74 6d 6f 73 69 63 20 42 54
@0014 11  00
@0018 12  00
@001c 18  04
@0020 01  11 e7 9b 36 ac d3
@0029 36  ff 00
@00c6 a0  ff 42 07 c4 c4 6f d1 ac 60 b4 e5 e0 9c 2b f8 bb
```

NVDS display with Verbose output for OTP NVDS:

```
C:\AtmosicMP\bin>nvds_tool.exe -i -r otp_nvds.bin -v
@0007 b1  0e db 07 2f d2 90 0f 00 fd 0d 00 00 87 e1 f0 03
@001a b0  21 00 00 00 e8 47 23 02 68 65 0f 02 b0 00 00 00 99 28 d1 10
@0031 bc  fc 72 05 a3 d5 62 15 a8 b7 78 41 06 fc 68 49 08 36 58 ad a3 00 00
00 00 99 d9 19 d4 00 00 00 00 b6 7d c1 de ea 6d 91 df b8 7e c1 e0 62 70 c1
e0 dd 7d 19 00 2b 6e 19 fe 68 7d 5d bd bf 6d 45 c0 b8 7e 01 00 62 70 01 00
05 7e 3d bd 41 6e 65 c0 00 00 38 bd ff ff ff ff 00 00 38 bd ff ff ff ff fc
10 1f 00 79 09 99 22 df dd 01 00 c6 18 23 00 00 04
@00a6 fc  09 2c 41 f8 6b 00
@00af b8  31 40
@00bd b5  00 b4 c4 04
@00c4 2e  f4 01
@0120 bd  18 03 00 00
@03cf 01  3c 5d 29 30 2c 70
@03d8 cc  eb ad 8f 00 99 0f 59 0d 88 0e fc f4 7e b8 fe ff 69 9a 90 8a 3c 5d
29 30 2c 70 83 ff
```

With the option -p , the output would be the readable string of each tag and data with address offset. The NVDS and OTP files with pretty output respectively are shown below:

NVDS display with pretty output for Flash/RRAM NVDS:

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin -p
@0007 DEVICE_NAME 'Atmosic BT'
@0014 SLEEP_ENABLE NONE
@0018 EXT_WAKEUP_ENABLE Disabled
@001c PROG_DELAY 4 half slots (1250us)
@0020 BD_ADDRESS D3:AC:36:9B:E7:11
@0029 RECENT_BOND 255
@00c6 LOC_IRK ff 42 07 c4 c4 6f d1 ac 60 b4 e5 e0 9c 2b f8 bb
```

NVDS display with pretty output for OTP NVDS:

```
C:\AtmosicMP\bin>nvds_tool.exe -i -r otp_nvds.bin -p
@0007 MDM_CAL AGCMEAS = [twomeg_det_min=(0x5e), twomeg_rssi_offset=(0x0f),
det_min=(0x5b), rssi_offset=(0x0e)]
              AGCGAIN = [pd_quick_drop=(0x3e), adc_quick_drop=(0x21), start=
(0x52)]
              DCOFF = [forced_q=(0x06), forced_i=(0xfe), force=(0x01)]
              IQCORR = [twomeg_phase_corr=(0x3f), twomeg_ampl_corr=(0x03),
twomeg_enable_iqcorr=(0x01), phase_corr=(0x03), ampl_corr=(0x03),
enable_iqcorr=(0x01)]

@001a RIF_CAL BIAS = [rbias=(0x21)]
              RXBBF_1M = [tia_rtrim=(0x02), tia_ctune=(0x23), rcc_rtrim=
(0x04), rc_ctune=(0x1f), notch_ctune=(0x28)]
              RXBBF_2M = [tia_rtrim=(0x02), tia_ctune=(0x0f), rcc_rtrim=
(0x06), rc_ctune=(0x15), notch_ctune=(0x28)]
              RXDC8 = [bb2i_dcoc=(0x05), bb2q_dcoc=(0x10)]
              SYNTX0 = [synth_spare=(0x00), cpLkIntLow=(0x01), loifdiv=
(0x00), txmodRelPathDelay=(0x01), txmodDelayHF=(0x01), txmodGain2M=(0x04),
txmodGain=(0x04), vcrefMi=(0x02), stateLength_CloseLoop_x2u=(0x04),
stateLength_CapSearchCount_x1u=(0x02), stateLength_CapSearchGetReady_x250n=
(0x03), stateLength_PowerOnWait_x4u=(0x01)]
PSEQ_XTAL_BITS1=0x1f10fc,PMU_TOP_PMU2A=0x22990979,PMU_TOP_PMU3=0x1dddf,PMU_TO
P_PMU4=0x2318c6,tx_gain_offset=0,misc=0x4
@00a6 PMU_W block=0x9,addr=0x2c,data=0x6b
@00af ATE  31 40
@00bd WATCHDOG 80000000 (in 16MHz cycles)
@00c4 SLEEP_ALGO_DUR 500 (250us)
@0120 CUST_CFG PSEQ_XTAL_BITS0=0x318,
@03cf BD_ADDRESS 70:2C:30:29:5D:3C
@03d8 cc  eb ad 8f 00 99 0f 59 0d 88 0e fc f4 7e b8 fe ff 69 9a 90 8a 3c 5d
29 30 2c 70 83 ff
```

14.2 Add, Erase Tag

The add tag option -t <tag> -d <data file > is used to replace or add tag. If the tag exists, the old tag will be erased and a new tag will be created at the end.

The BD_ADDRESS (0x01) tag is an example to demonstrate how to assign a new Bluetooth address 11:22:33:44:55:66 through this option.

1) Apply -i -r otp_nvds.nvm -p to show the otp_nvds.nvm OTP file. The output contains BD_ADDRESS with 7C:50:43:3E:65:9A content.

otp_nvds.nvm content:

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin -p
@0007 DEVICE_NAME 'Atmosic BT'
@0014 SLEEP_ENABLE NONE
@0018 EXT_WAKEUP_ENABLE Disabled
@001c PROG_DELAY 4 half slots (1250us)
@0020 BD_ADDRESS D3:AC:36:9B:E7:11
@0029 RECENT_BOND 255
@00c6 LOC_IRK ff 42 07 c4 c4 6f d1 ac 60 b4 e5 e0 9c 2b f8 bb
```

BD Address is at 0x0020

2) Prepare a binary file for data of the new BD_ADDRESS tag.

```
00000000  66 55 44 33 22 11
```

3) Add -t 01 -d bdaddr.bin on the command line and see the address changed to 11:22:33:44:55:66.

CONFIDENTIAL

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin -p -t 01 -d bd_addr.bin
@0007 DEVICE_NAME 'Atmosic BT'
@0014 SLEEP_ENABLE NONE
@0018 EXT_WAKEUP_ENABLE Disabled
@001c PROG_DELAY 4 half slots (1250us)
@0029 RECENT_BOND 255
@00c6 LOC_IRK ff 42 07 c4 c4 6f d1 ac 60 b4 e5 e0 9c 2b f8 bb
@00d9 BD_ADDRESS 11:22:33:44:55:66
```

BD Address is moved to 0x00D9

The erase tag operation marks the tag invalid. If the command line of the add tag example is modified from -t 01 -d bdaddr.bin to -e 01 , the BD_ADDRESS tag will not show on the output.

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin -p -e 01
@0007 DEVICE_NAME 'Atmosic BT'
@0014 SLEEP_ENABLE NONE
@0018 EXT_WAKEUP_ENABLE Disabled
@001c PROG_DELAY 4 half slots (1250us)
@0029 RECENT_BOND 255
@00c6 LOC_IRK ff 42 07 c4 c4 6f d1 ac 60 b4 e5 e0 9c 2b f8 bb
```

Tag BD Address has been removed

For more information on tag ID, please refer to section Common OTP/NVDS/MP Region Tag ID.

CONFIDENTIAL

14.3 Generate New OTP/NVDS/MP Region File

The -o <file> option is used for generating new OTP or NVDS/MP region files. It should be used with the binary output -b option. In the previous add tag example, If change the -p to -b -o otp_nvds_new.nvm , then a new file called otp_nvds_new.nvm will be generated.

Generate a new OTP File:

```
C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds.bin -b -o flash_nvds_new.bin -t
01 -d bd_addr.bin

C:\AtmosicMP\bin>nvds_tool.exe -r flash_nvds_new.bin -p
@0007 DEVICE_NAME 'Atmosic BT'
@0014 SLEEP_ENABLE NONE
@0018 EXT_WAKEUP_ENABLE Disabled
@001c PROG_DELAY 4 half slots (1250us)
@0029 RECENT_BOND 255
@00c6 LOC_IRK ff 42 07 c4 c4 6f d1 ac 60 b4 e5 e0 9c 2b f8 bb
@00d9 BD_ADDRESS 11:22:33:44:55:66
```

CONFIDENTIAL

15. Common OTP/NVDS/MP Region Tag ID

In sections OTP/NVDS/MP Tag Addition with File and Generate New OTP/NVDS/MP Region File, Tag ID can refer to Table 15-1.

NVDS / MP Region and OTP	
BD address	0x01
NVDS	
Device name	0x02
MP Region	
CAP	0xAE
OTP	
CAP	In OTP data, CAP value is a part of Tag 0xBD and is not independent bytes, it is recommended to use Set 16M Crystal CAP to modify the CAP value.

Table 15-1 NVDS/MP Region and OTP Tag ID

16. Troubleshooting

1) Figure 16-1 shows error log No board found



Figure 16-1 No Board Found

Please ensure the USB cable is connected. Open the device manager to check if libusbK USB Devices exist. If not, please refer to section Install MFx Interface to install USB driver.

2) Figure 16-2 shows error log Different type of boards found simultaneously .

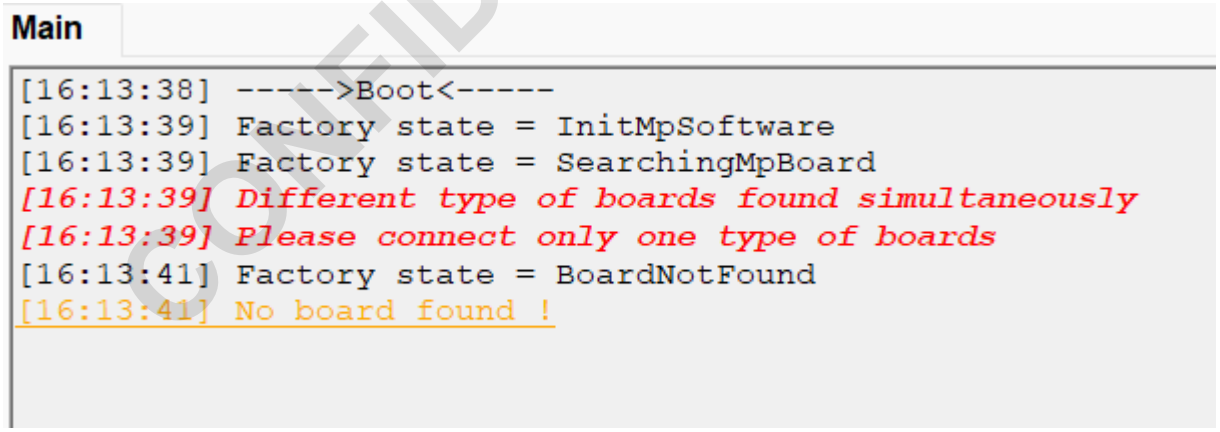


Figure 16-2 Different Type of Boards Found Simultaneously Error Log

Please refer to the Install MFx Interface section to install USB drivers.

3) Figure 16-3 shows Invalid board <num> found and the num is bigger than 32768.

CONFIDENTIAL

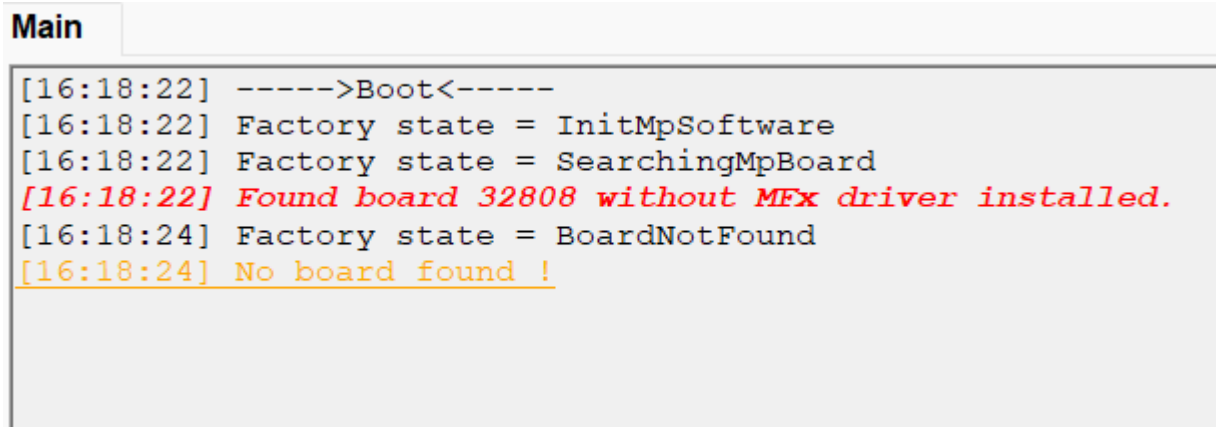


Figure 16-3 Invalid Board <num> Found

Please refer to the section Install MFx Interface to install USB drivers.

CONFIDENTIAL

Revision History

Date	Version	Description
March 17, 2023	0.59	Updated for MPTool version 2.1.8.1: Added Hardware and Software Requirements, DUT Design Requirements for Manufacturing sections, Figure 10-34 PV Test Setup. Updated Select ATM File, Choose Testing Options, Help Menu, Run Tab, Crystal Trim, RX Sensitivity, User Firmware (.atm, .mpbin) / NVDS file (.bin) / Tags Customization, OTP File (.nvm), Write MP Region, PV, Harvesting Test, Calibration settings, Atmosic Production Tool Console Mode, DISPLAY Content of NVDS and OTP file.
November 11, 2022	0.58	Added support for MP Tool v2.1.7.3 and ATM33/e. Added Photovoltaics (PV) Test, External NVDS process, Update Test Result with External Process sections. Updated Overview, Crystal Trim, User Firmware (.atm, .elf) / NVDS file (.bin) / Tags Customization, UART0 Pin Modification, Command Mode, DUT Firmware Programming, DUT Get PV count sections.
September 15, 2022	0.57	Added support for MP Tool v2.1.0.0. Added DL Board Versions, Uninstall any previous versions of the MPTool, GUI Mode Quick Start, Set 16M Crystal CAP, FTDI pin control, DUT RRAM Firmware, Programming, DUT Secure Journal dump or push, DUT RRAM Erase sections. Updated Install MFx Interface, Set 16M Crystal CAP, DUT RRAM Firmware Programming sections, Figure 6-2 USB Serial Port Properties, Figure 6-3 Advanced Setting for COM ports,Figure 7-2 Load NVDS, Figure 7-3 Choose Test Options, Figure 7-4 Burn_RAM Process, Figure 8-1 Atmosic Mass Product Tool User Interface, Run Tab, Settings Tab, Crystal Trim, Frequency Test sections, changed format, corrected typos.
December 14, 2021	0.56	Format change. Added Dump flash to bin file, Flash chip erase, Check protecting, Show ATM information, Export FW bin file from ATM file, Export NVDS bin file from ATM file sections.
November 10, 2021	0.55	Updated section NVDS/MP Region Create Blank File. Added sections System Environment Variable, Pull MP region and Set UART baud rate setting.
October 25, 2021	0.54	Updated Write MP Region, Setting Tab, Figure 19 -

CONFIDENTIAL

		Run Tab Before Testing, Figure 20 - Run Tab Under Testing, Figure 21- Run Tab After Testing Success, Figure 22 - Run Tab After Testing Fail, Figure 23 - Setting Tab, Figure 24 - Crystal Trim Setting, Figure 34 - User Firmware/Tags Setting, Figure 39 - Start Option, added Figure 41 - MP Region Information, Figure 42 - Write Calibration Data to MP Region, Figure 43 - Choose Which Calibration Data to Write, Figure 64 - CLI Command Detail Logs, Set Testing Log Refresh, Set Testing Log Finished, Write MP region, Multiple DUT Testing in CLI Mode, Common OTP/NVDS/MP Region Tag ID, updated NVDS Tool,
September 30, 2021	0.53	Updated Figure 8 - Start Menu Shortcuts, Figure 19 - Run Tab Before Testing
July 23, 2021	0.52	Format change; updated Setting Menu, Figure 14 - MP Tool Setting, Help Menu, Figure 18 - MP Tool Help Menu, Figure 20 - Setting Tab, Figure 34 - User Firmware/Tags Setting, User Firmware (.elf)/NVDS file (.bin)/Tags Customization; added sections Environment Setup for DL, Adjust Serial Port Latency Timer, UART0 Pin Modification, Start Option, Atmosic Production Tool Console Mode, NVDS Tool
May 11, 2020	0.51	Added sections OTP file (.nvm), Troubleshooting.
March 25, 2020	0.50	Initial version created.

CONFIDENTIAL

ATMOSIC TECHNOLOGIES – DISCLAIMER

This product document is intended to be a general informational aid and not a substitute for any literature or labeling accompanying your purchase of the Atmosic product. Atmosic reserves the right to amend its product literature at any time without notice and for any reason, including to improve product design or function. While Atmosic strives to make its documents accurate and current, Atmosic makes no warranty or representation that the information contained in this document is completely accurate, and Atmosic hereby disclaims (i) any and all liability for any errors or inaccuracies contained in any document or in any other product literature and any damages or lost profits resulting therefrom; (ii) any and all liability and responsibility for any action you take or fail to take based on the information contained in this document; and (iii) any and all implied warranties which may attach to this document, including warranties of fitness for particular purpose, non-infringement and merchantability. Consequently, you assume all risk in your use of this document, the Atmosic product, and in any action you take or fail to take based upon the information in this document. Any statements in this document in regard to the suitability of an Atmosic product for certain types of applications are based on Atmosic's general knowledge of typical requirements in generic applications and are not binding statements about the suitability of Atmosic products for any particular application. It is your responsibility as the customer to validate that a particular Atmosic product is suitable for use in a particular application. All content in this document is proprietary, copyrighted, and owned or licensed by Atmosic, and any unauthorized use of content or trademarks contained herein is strictly prohibited.

Copyright ©2020-2023 by Atmosic Technologies. All rights reserved. Atmosic logo is a registered trademark of Atmosic Technologies Inc. All other trademarks are the properties of their respective holders.

CONFIDENTIAL

Atmosic™

Atmosic Technologies | 2105 S. Bascom Ave. | Campbell CA, 95008
www.atmosic.com